

SOFTWARE REQUIREMENT SPECIFICATION

Aura Moon (HOS-03)

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I. Record of Changes

Date	A* M, D	In charge	Change Description

*A - Added M - Modified D - Deleted

II. Software Requirement Specification

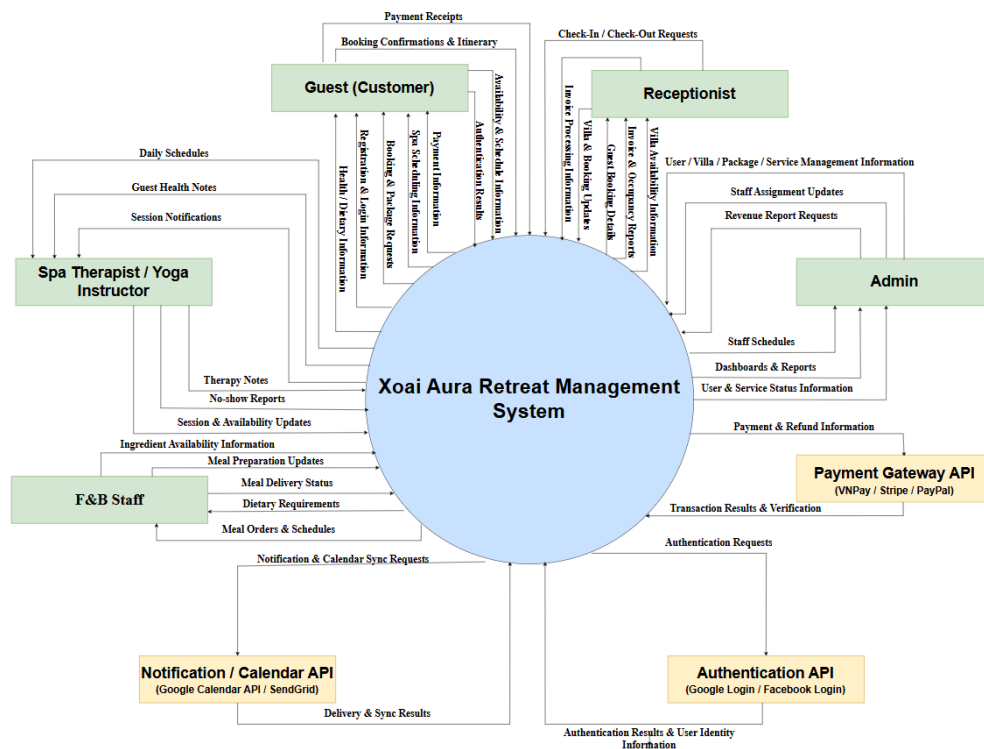
1. Overall Requirements

1.1 Context Diagram - My

The Xoai Aura Retreat Management System is a wellness resort management platform designed to support retreat package booking, villa accommodation management, spa scheduling, dietary planning, billing, and business reporting.

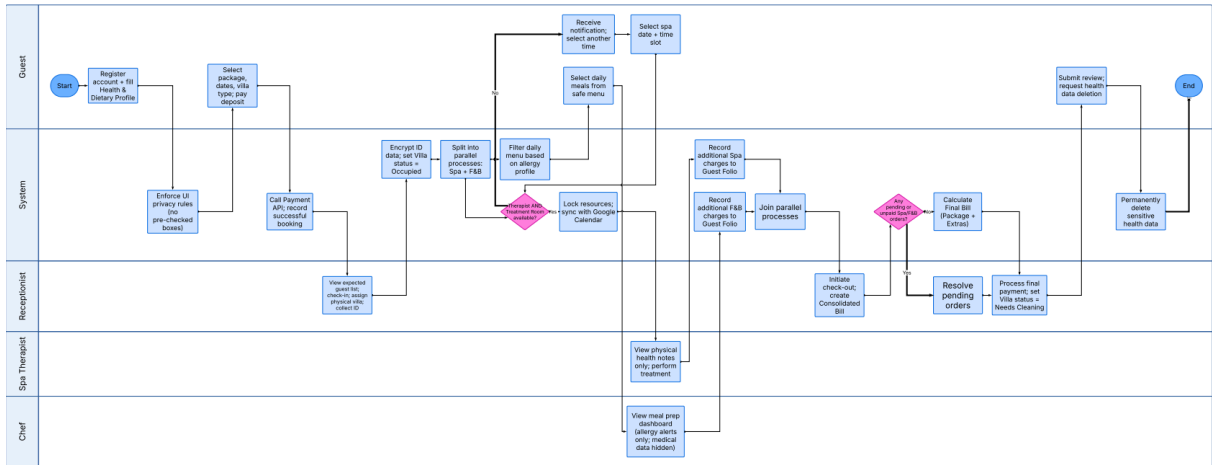
The context diagram below illustrates the system boundary and its interactions with external entities, including Guests, Receptionists, Spa Therapists/Yoga Instructors, F&B Staff, Administrators, and Managers. The system also integrates with external services such as Payment Gateway APIs, Authentication APIs, and Notification & Calendar APIs.

These interactions enable the efficient management of retreat operations, guest services, payment processing, and business analytics.



Context Diagram

1.2 Main Business Processes - Hải



Main Business Processes

Step #	Step Name	Detailed Description	Role	Note
1	Publish driving course enrollment	1. Activity: - Guest creates an account and fills out the "Health & Dietary Profile" (medical conditions, allergies). - System checks the UI to ensure NO consent checkboxes are pre-checked. 2. Input: Email, Password, Sensitive medical/allergy data. 3. Output: User account, Health profile (encrypted in DB).	Guest, System	Crucial: Must strictly comply with Decree 356/2025 regarding sensitive data.

2	Retreat Package Booking & Deposit	1. Activity: - Guest filters and selects a Retreat Package by goal, chooses dates and Villa type. - Redirects to Payment Gateway (Stripe/VNPay) to pay the deposit. 2. Input: Retreat_Package_ID, Arrival/Departure Dates, Villa Type, Card details. 3. Output: Booking Record, Successful deposit payment status.	Guest, System	When clicking the "register for course" button, user authentication is required
3	Check-in & Room Assignment	1. Activity: - Receptionist views the expected guest list, assigns a physical Villa number. - Collects guest's ID/CCCD info for temporary residence declaration. - System encrypts the ID and changes room status. 2. Input: Booking ID, ID/Passport details, Physical Villa number. 3. Output: Villa Status: "Occupied", ID data securely stored.	Receptionist, System	Complies with Residence Law 2020. Receptionist MUST NOT view the guest's medical data.

4	Spa Scheduling	1. Activity: - Guest selects date/time for a Spa session. - System runs a DB locking transaction: Finds ONE available therapist AND ONE available room simultaneously. If available, confirms the schedule and calls Google Calendar API to notify the guest. 2. Input: Booking ID, Spa_Service_ID, Desired time slot. 3. Output: Confirmed Spa schedule, Reminder email (sent 1h prior).	Guest, System	Complex Logic: 2-Dimensional Double-Booking prevention constraint
5	Spa Treatment Execution	1. Activity: - Therapist views their work schedule and reads physical health notes (other data is hidden). - Performs treatment and updates the session status. - System automatically accumulates charges (if the guest uses extra off-package services) into the Guest Folio. 2. Input: Spa Schedule, Physical therapy notes. 3. Output: Status: "Completed/No-Show", Updated Guest Folio.	Spa Therapist, System	RBAC: Therapist MUST NOT view the guest's dietary/allergy information.

6	F&B Meal Ordering	1. Activity: - System cross-references the "Dietary Profile" to filter out menu items containing allergens. - Guest selects daily meals from this pre-filtered safe menu. 2. Input: Original Menu, Guest's allergy data. 3. Output: Safe Menu, F&B Order.	Guest, System	Applies the Data Minimization principle
7	Culinary Preparation	1. Activity: - Chef opens the Dashboard to view aggregated meal orders and "allergy alerts". - Chef prepares the dish and updates the status (Preparing -> Ready to deliver). 2. Input: F&B Order, Food allergy alerts. 3. Output: Updated meal status, A-la-carte charges billed to Guest Folio.	Chef / F&B, System	RBAC: Chef MUST NOT view the guest's physical medical records

8	Check-out & Consolidated Bill	1. Activity: - Receptionist clicks Check-out. System scans to check if there are any pending/unpaid Spa/F&B orders. - If clear, the system aggregates: Remaining Package fee + Extra services into 1 Consolidated Bill. - Receptionist collects payment and releases the room. 2. Input: Guest Folio (Room + Spa + F&B). 3. Output: Final Bill, Villa Status: "Needs Cleaning".	Receptionist, System	Constraint: Guest CANNOT check out if there are pending orders.
9	Review & Data Deletion	1. Activity: - Guest submits a Retreat quality review form. - Guest exercises their "Right to be forgotten". System executes a command to permanently wipe medical and allergy records from the DB. 2. Input: Rating/Review, Data deletion request. 3. Output: System review record, User profile "cleansed" of sensitive data	Guest, System	Ensures absolute privacy after the retreat concludes.

1.3 User Requirements - Đặc

1.3.1 Actors

[An actor is someone/something that interacts with the system.]

- *The only external entities that interact with the system*
- *Actors are outside the system and not part of it*

System Administration

- A user is an individual, whereas an actor represents the role played by all users of the same type
- There are other types of actors in addition to or in place of human actors: external systems, I/O devices, or timers

Following are some questions you might ask to help user representatives identify actors

- Who (or what) is notified when something occurs within the system?
- Who (or what) provides information or services to the system?
- Who (or what) helps the system respond to and complete a task?

This part gives the description of system actors, you can follow the table form as below]

#	Actor	Description
1	Guest / Customer	A customer who uses the system to register, log in, complete health and dietary profiles, browse wellness packages, book villas, schedule spa/treatment sessions, pre-select meals, view itinerary, make payments, and submit reviews.
2	Receptionist	A front-desk staff member who manages guest check-in/check-out, assigns physical villas, manages villa status, books additional spa services for guests, and processes consolidated invoices and final payments.
3	Spa Therapist / Yoga Trainer	A service provider who views daily treatment schedules, checks treatment-relevant health notes, and updates treatment session status such as Completed or No-Show.
4	Chef / F&B Staff	A food and beverage staff member who views daily meal preparation dashboards, checks food allergy alerts, prepares personalized meals, and updates meal order status.
5	Administrator	A system administrator who manages staff accounts, assigns user roles, configures role-based access control, and manages master data such as villa types, spa services, retreat packages, and staff records.
6	Resort Manager	A management user who monitors revenue dashboards, reviews business performance, and exports monthly reports on room occupancy and therapist utilization.
7	Payment Gateway	An external payment service such as Stripe, VNPay, or PayPal that processes deposit payments and final payments securely.
8	Calendar / Notification Service	An external service such as Google Calendar or SendGrid that synchronizes spa/yoga schedules and sends reminder notifications to guests.
9	SSO Provider	An external authentication provider such as Google Identity or Facebook Login that supports secure single sign-on for guests.

System Administration

1.3.2 Use Cases (UC)

[A use case (UC) describes a sequence of interactions between a system and an external actor that results in the actor being able to achieve some outcome of value. The names of use cases are always written in the form of a verb followed by an object. Select strong, descriptive names to make it evident from the name that the use case will deliver something valuable for some user.]

Following are some questions you might ask to help user representatives identify use cases

- *What will the actor use the system for?*
- *Will the actor create, store, change, remove, or read data in the system?*
- *Will the actor need to inform the system about external events or changes?*
- *Will the actor need to be informed about certain occurrences in the system?*

This part describes the use cases you could define, you can follow the table form as below]

ID	Use Case	Feature	Use Case Description
01	Register, Verify Email and Log In	Authentication1	The Guest creates an account, verifies email, and logs in securely using system credentials or SSO.
02	Complete Health and Dietary Profile	Health Profile Management	The Guest provides health conditions, allergies, and dietary preferences with explicit consent.
03	Manage Staff Accounts and Assign Roles	User and Role Management	The Administrator creates staff accounts and assigns strict roles such as Therapist, Chef, and Receptionist.
04	Manage Master Data	Master Data Management	The Administrator manages villa types, spa services, retreat packages, and staff records.
05	Delete Sensitive Health Data	Data Privacy Management	The Guest requests permanent deletion of sensitive health and allergy data after the stay.
06	Browse Wellness Packages	Package Browsing	The Guest browses available wellness packages and filters them by goals such as Detoxification, Yoga, Stress Relief, or Weight Loss.
07	Book Wellness Package and Pay Deposit	Booking and Payment	The Guest selects a package, arrival date, villa type, and pays a secure deposit.
08	Check In Guest	Reception Management	The Receptionist views expected arrivals, checks in guests, assigns a specific villa, and collects required identity information.
09	Manage Villa Status	Villa Management	The Receptionist updates villa status such as Available, Occupied, Under Maintenance, or Cleaning Required.

10	View Booking Details and Itinerary Timeline	Booking Tracking	The Guest views booking details, villa information, spa schedule, meal plan, and billing timeline.
11	Schedule Spa/Treatment Session	Spa Scheduling	The Guest schedules included spa or therapy sessions by selecting a date and time slot.
12	Find Available Therapist and Treatment Room	Automatic Scheduling	The System automatically checks both therapist availability and treatment room availability before confirming a session.
13	View Daily Work Schedule	Therapist Schedule Management	The Spa Therapist or Yoga Trainer views assigned daily sessions and treatment-relevant health notes.
14	Update Treatment Session Status	Treatment Management	The Spa Therapist marks a session as Completed or No-Show.
15	Book Additional Spa Service	Additional Service Booking	The Receptionist manually books extra spa services for guests and posts the charge to the guest folio.
16	Pre-select Daily Meals	Meal Management	The Guest selects daily meals from a personalized menu filtered by allergies and dietary restrictions.
17	View Daily Meal Preparation Dashboard	F&B Dashboard	The Chef or F&B Staff views aggregated meal orders and relevant food allergy alerts for the selected date.
18	Update Meal Order Status	Meal Order Management	The Chef updates meal order status from Preparing to Ready for Delivery.
19	Order A-la-carte Food and Beverage	Additional F&B Service	The Guest orders additional food or beverages outside the package and the system posts the charge to the folio.
20	Enforce Data Minimization for F&B Staff	Data Privacy and RBAC	The System restricts kitchen staff from viewing medical history and only displays food allergies and dietary restrictions.
21	Generate Consolidated Invoice	Billing Management	The Receptionist generates a consolidated invoice including remaining package charges, additional spa services, and F&B orders.
22	Process Final Payment and Complete Check-out	Checkout and Payment	The Receptionist processes the final payment and updates villa status after successful check-out.

System Administration

23	Submit Post-stay Review and Rating	Feedback Management	The Guest submits a review and rating after completing the stay.
24	View Revenue Analytics Dashboard	Analytics and Reporting	The Resort Manager views revenue charts categorized by package, spa, and F&B income.
25	Export Monthly Occupancy and Therapist Utilization Report	Report Export	The Resort Manager exports monthly reports on room occupancy and therapist utilization to Excel.

1.3.2 Use Case Diagrams

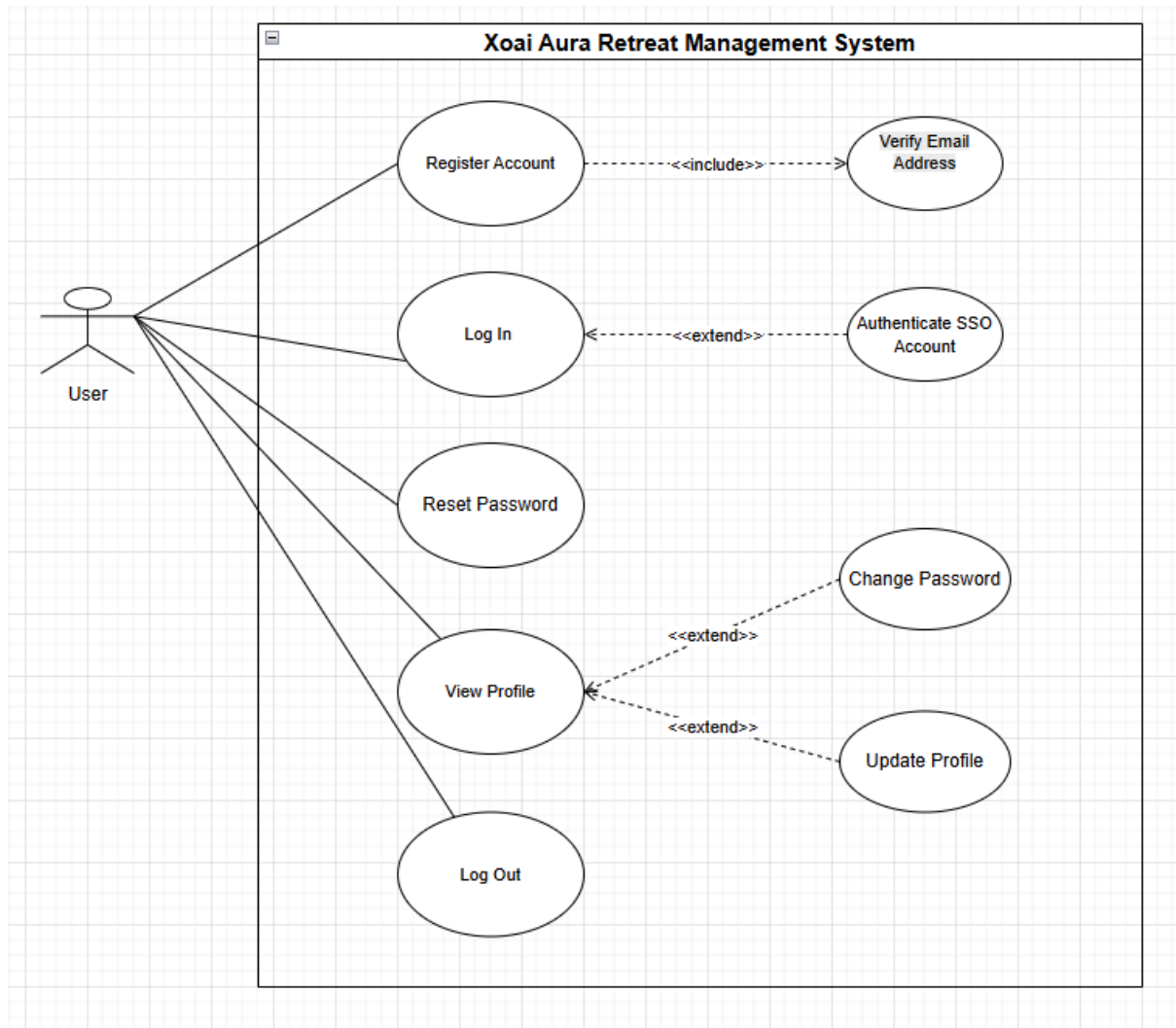
In this section, you need to provide the UC diagram(s) to show the actor-UCs and UC-UC relationships like the sample below. You can have multiple UC diagrams for the system, each diagram is for one actor or one workflow]

1.3.2.1 UCs for Guest



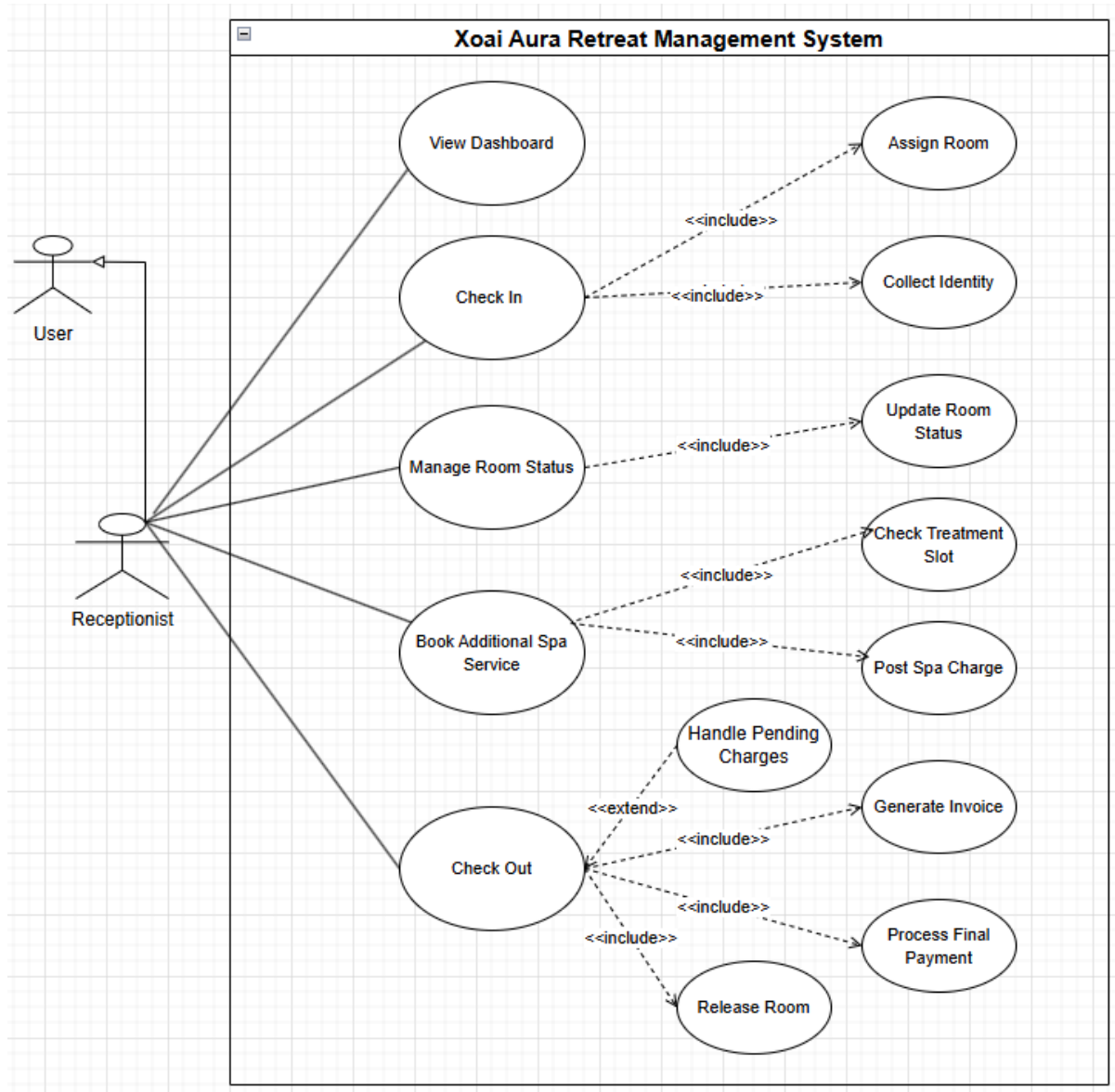
System Administration

1.3.2.2 UCs for User



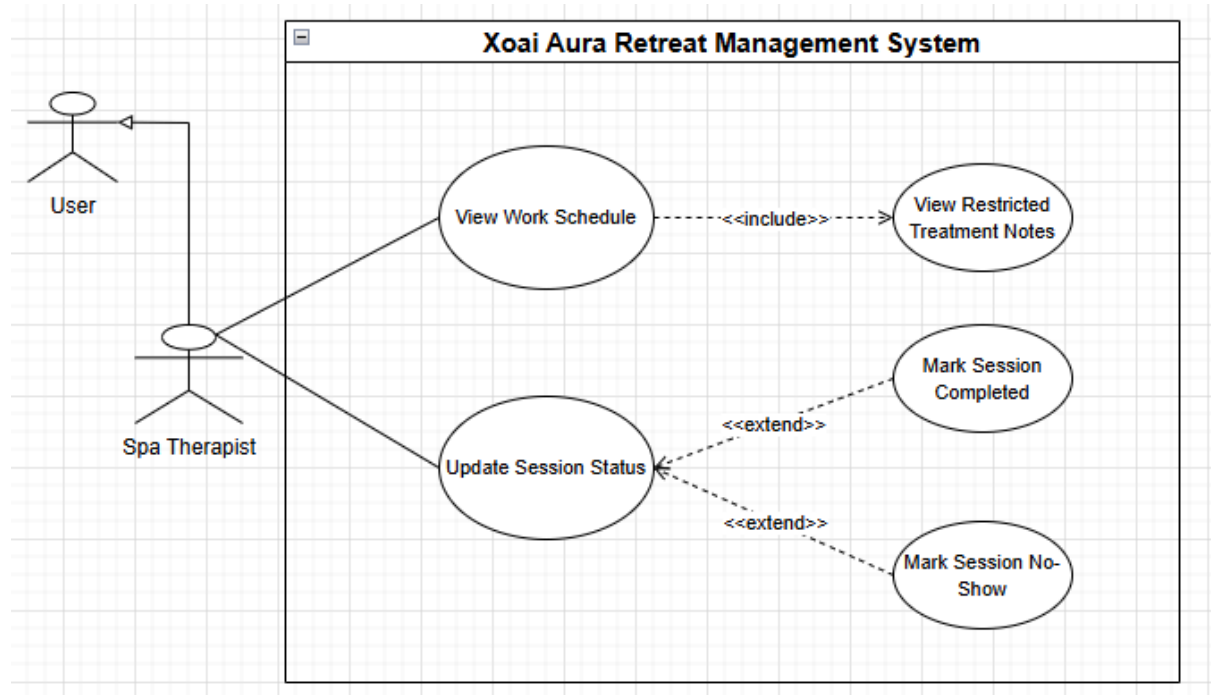
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1.3.2.3 UCs for Receptionist

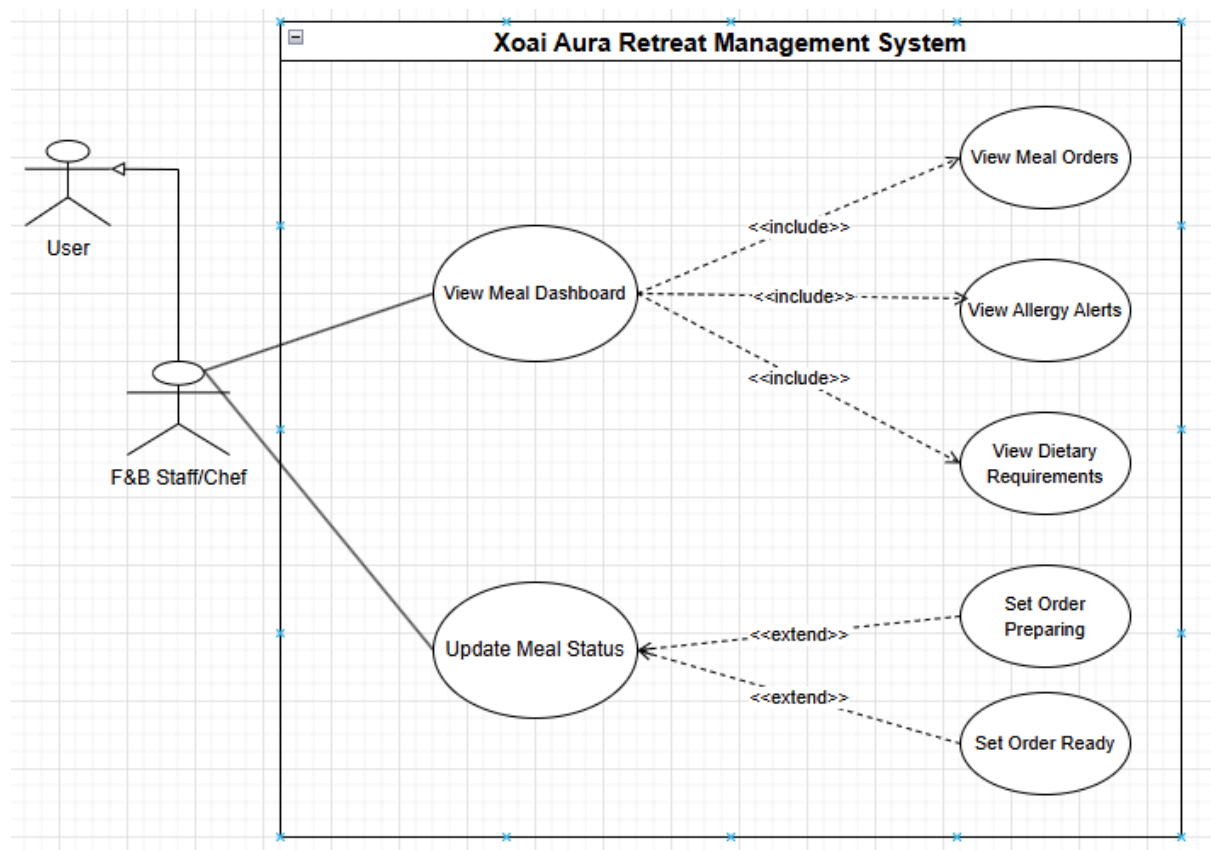


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1.3.2.4 UCs for Spa Therapist / Yoga Trainer

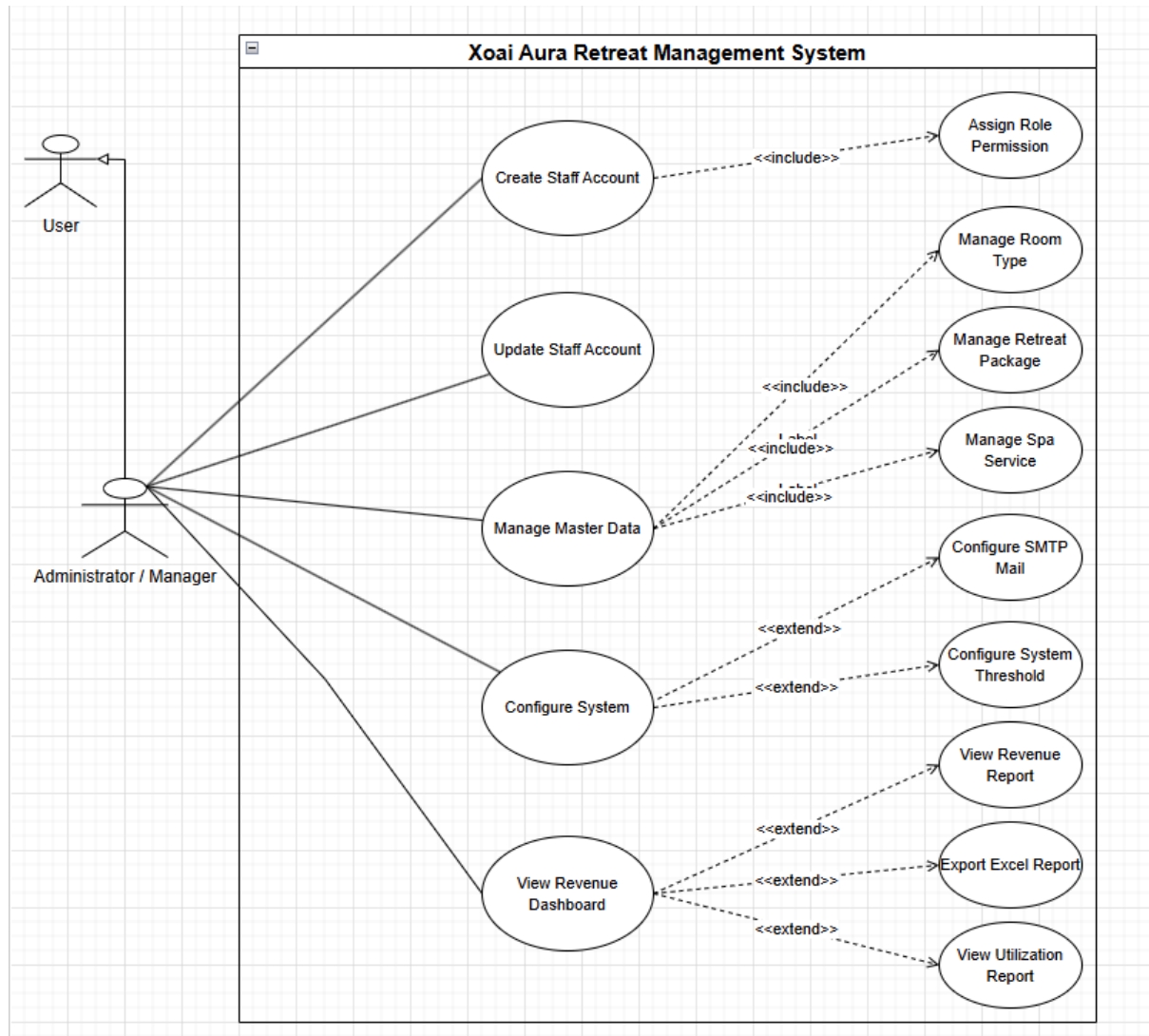


1.3.2.5 UCs for F&B Staff / Chef



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1.3.2.6 UCs for Administrator / Manager



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1.4 System Functionalities - DƯƠng

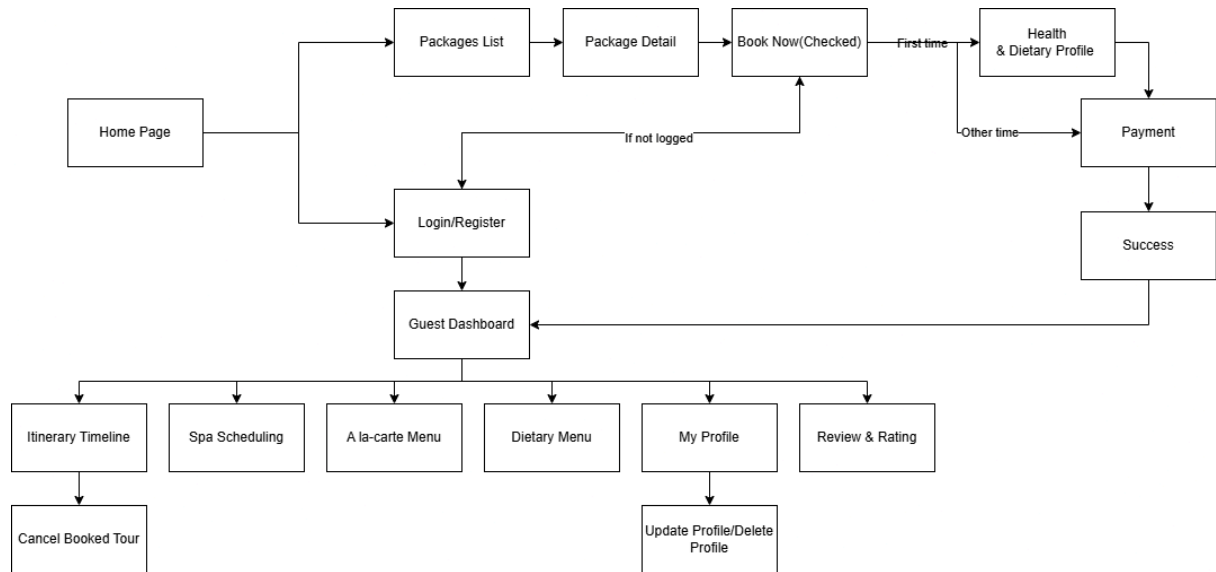
[Provide functionality overview of software system: screen flow, screen descriptions, system user roles, screen authorization, non-screen functions, ERD]

1.4.1 Screens Flow

[This part shows the system screens and the relationship among screens. You can draw the Screens Flow for the system in the form of diagram as below.]

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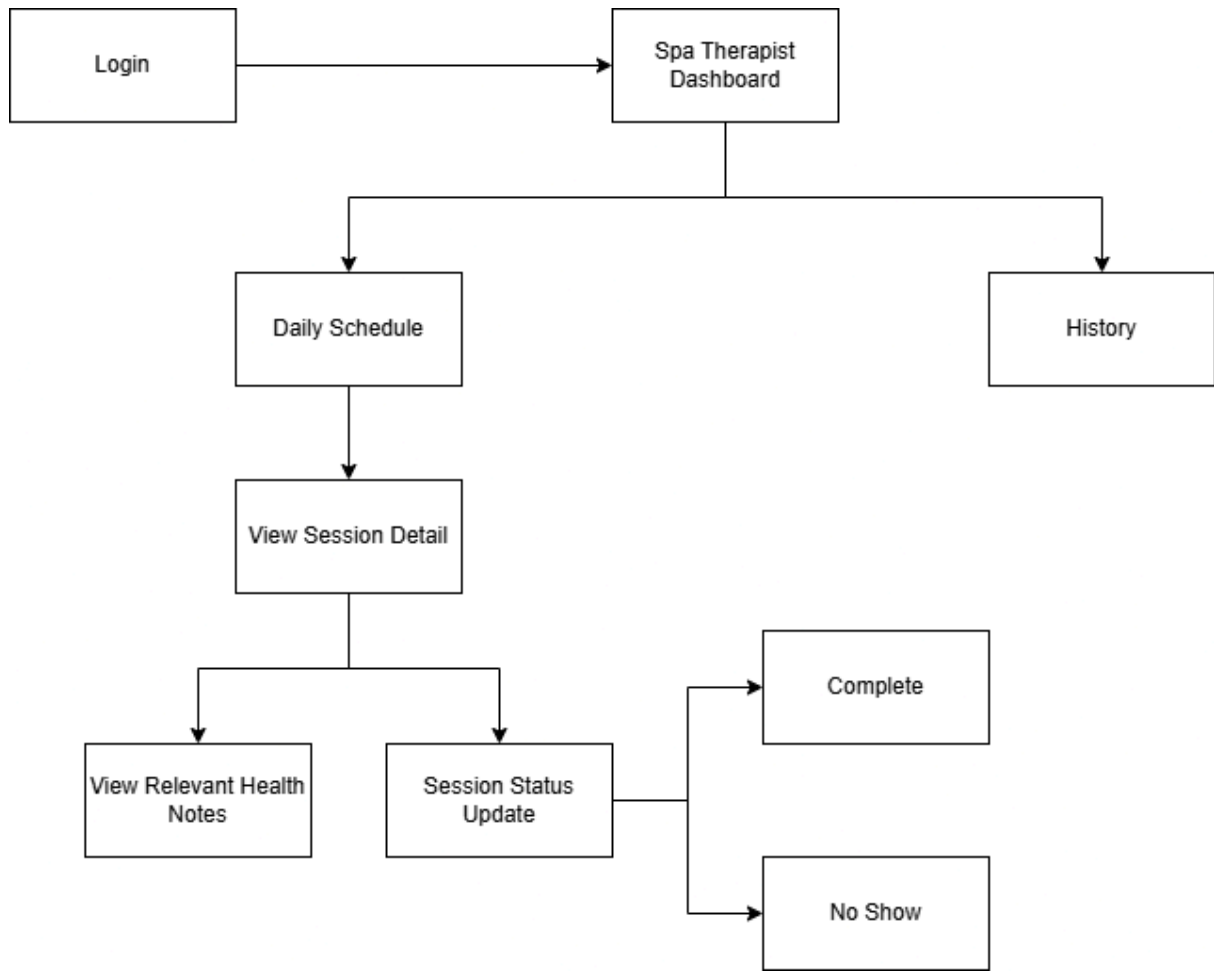
1.4.1.1 User Screen



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1.4.1.2 Spa Therapist Screen

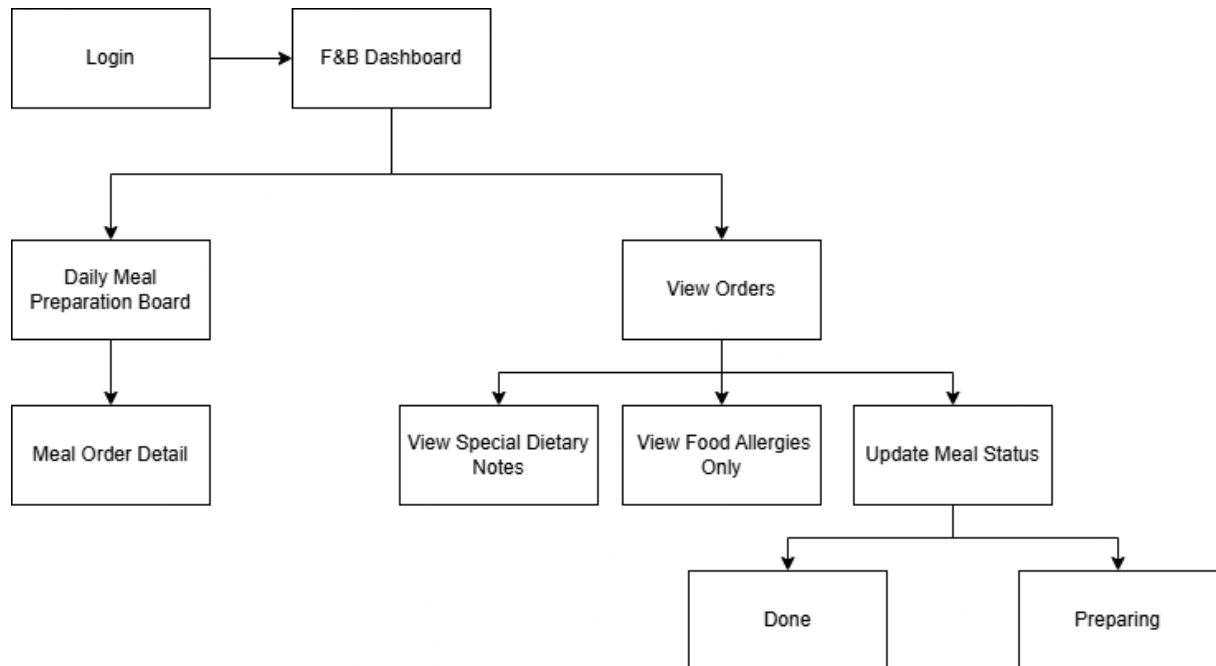
System Administration



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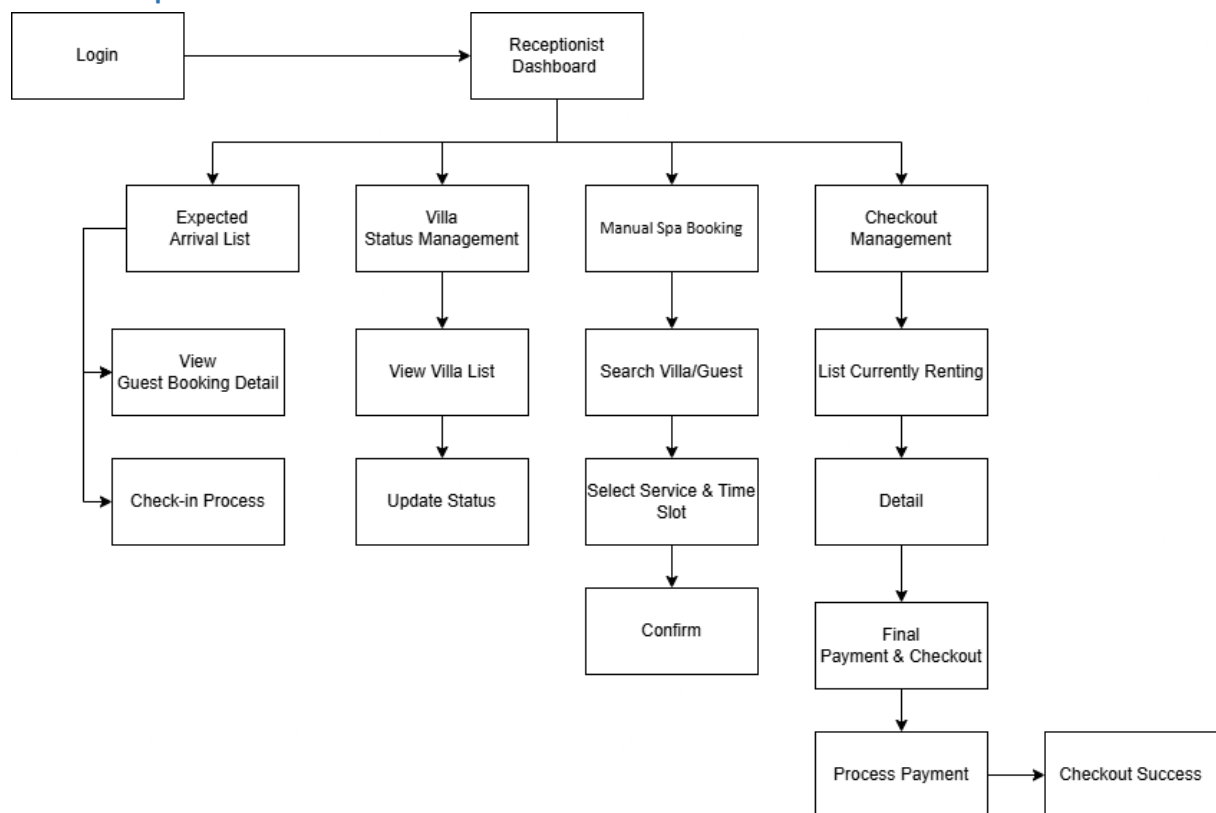
System Administration

1.4.1.3 F&B Screen



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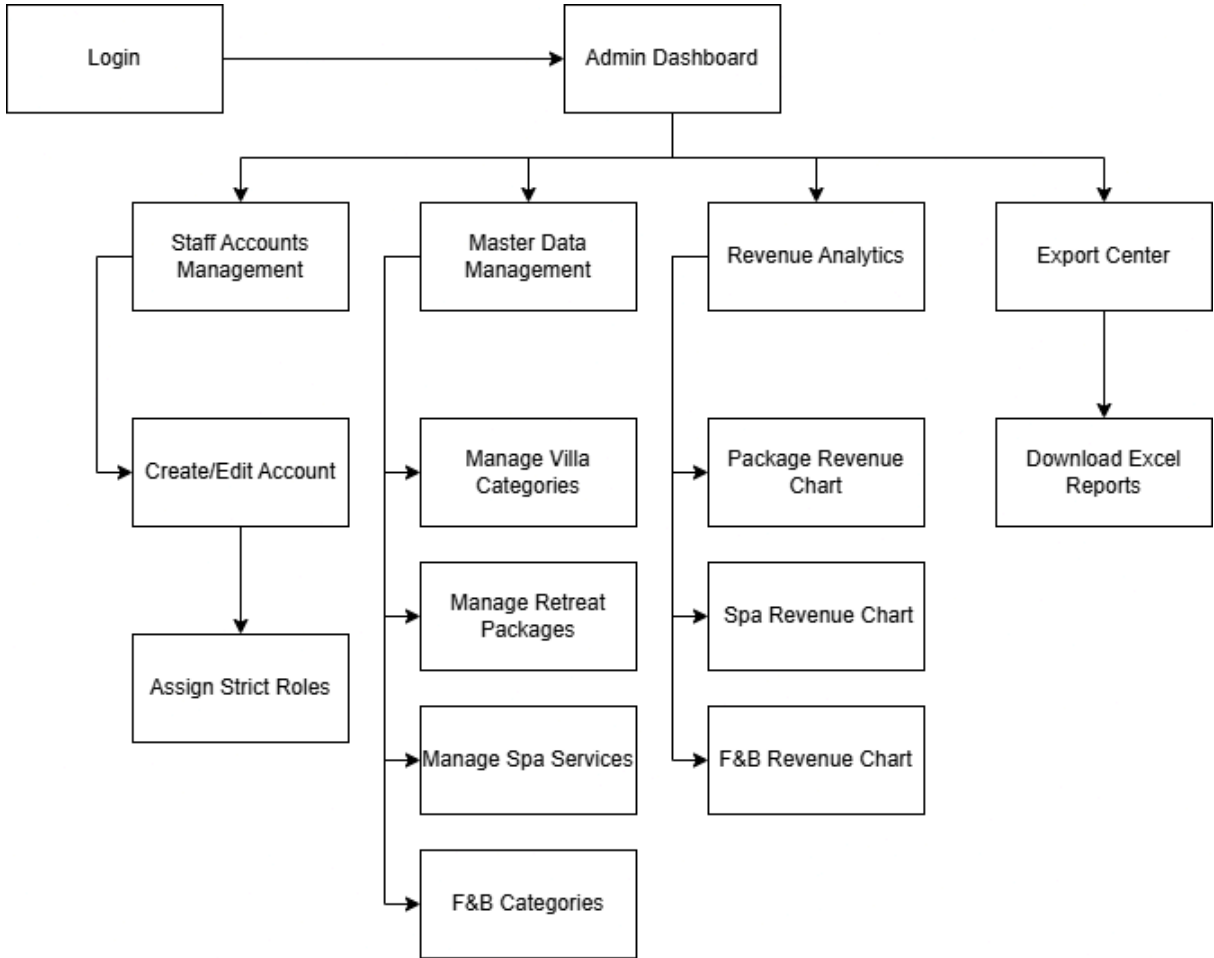
1.4.1.4 Receptionist Screen



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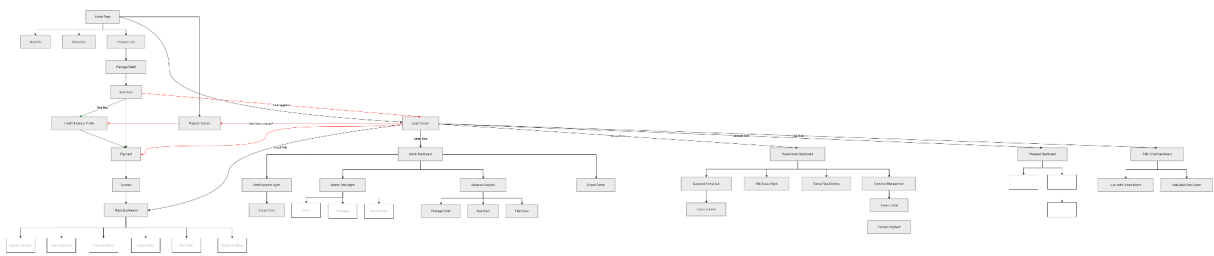
System Administration

1.4.1.5 Admin Screen



[Link](#)

1.4.1.6 All System



[Screen Flow](#)

1.4.2 Screen Authorization

[Provide the system roles authorization to the system features (down to screens, and event to the screen activities if applicable) in the table form as below – replace Role-Name1, Role-Name2, ... with your specific system user role names]

System Administration

Screen	Guest	Receptionist	Therapist	F&B / Chef	Admin
Home Page	X	X	X	X	X
About Us	X	X	X	X	X
Contact Us	X	X	X	X	X
Packages List	X	X	X	X	X
Package Detail	X	X	X	X	X
Login Screen	X	X	X	X	X
Register Screen	X				
Book Now [Action]	X				
Health & Dietary Profile	X				
Payment	X				

System Administration

Success Page	X				
Guest Dashboard	X				
Itinerary Timeline	X				
Spa Scheduling	X				
A la-carte Menu	X				
Dietary Menu	X				
My Profile	X				
Update/Delete Profile	X				
Review & Rating	X				
Receptionist Dashboard		X			
Expected Arrival List		X			

System Administration

Check-in Form		X			
Villa Status Management		X			
Update Villa Status [Action]		X			
Manual Spa Booking [Modal]		X			
Checkout Management		X			
Invoice Detail		X			
Process Payment		X			
Therapist Dashboard			X		
Daily Schedule			X		
Session Detail			X		

System Administration

View Health Notes [Modal]			X		
Update Session Status [Action]			X		
History			X		
F&B / Chef Dashboard				X	
Daily Meal Prep Board				X	
A-la-carte Orders Board				X	
Update Prep Status [Action]				X	
Admin Dashboard					X
Staff Accounts Management					X
Assign Roles [Modal]					X

System Administration

Master Data Management					X
Villas Management					X
Packages Management					X
Spa Services Management					X
Revenue Analytics					X
Package Revenue Chart					X
Spa Revenue Chart					X
F&B Revenue Chart					X
Export Center					X
Daily ID Report					X
Excel Reports					X

System Administration

1.4.3 Non-UI Functions

[Provide the descriptions for the non-screen system functions, i.e batch/cron job, service, API, etc.]

#	Feature	System Function	Description
1	Authentication	Google OAuth API	Allows users to sign in using Google accounts.
2	Payment	VNPay Payment Gateway API	Processes online payments through VNPay.
3	Payment	Payment Verification Service	Verifies transaction status and updates booking records.
4	Data Management	Database Backup Service	Performs scheduled database backups.

1.5 Entity Relationship Diagram - NgQC

1.5.1 Entity Relationship Diagram

[the ERD using the Crow-Foot notation]

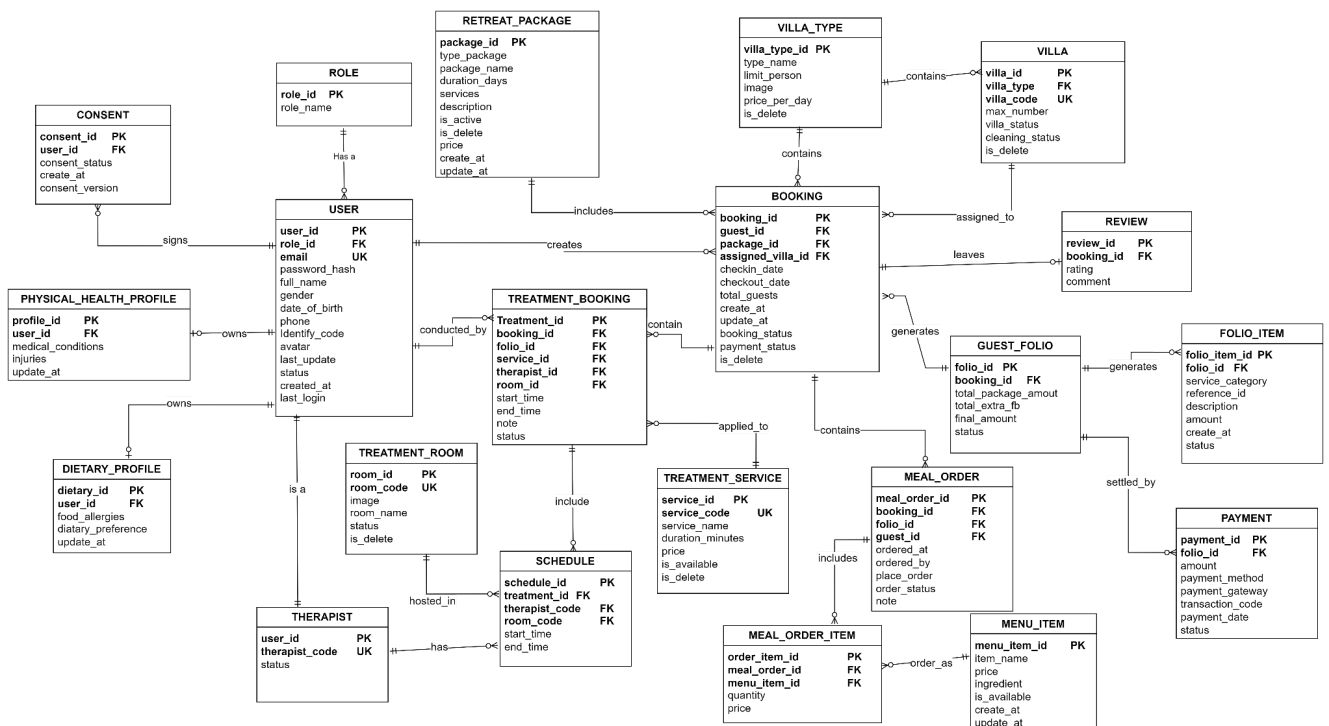


Figure x - ERD

1.5.2 Entities Description

#	Entity	Description
1	User	Stores personal information and account details of users in the system.
2	Role	Defines user roles (e.g., guest, staff, administrator).
3	Consent	Stores the status and version of user consent.
4	Physical_Health_Profile	Stores physical health information of users, such as medical conditions and injuries.
5	Dietary_Profile	Stores information regarding dietary preferences and food allergies of users.
6	Retreat_Package	Contains the catalog of retreat packages offered by the facility.
7	Villa_Type	Categorizes villa types, including capacity and pricing information.
8	Villa	Lists specific villa units associated with different types.
9	Booking	Manages guest reservation details.
10	Review	Stores feedback and ratings provided by guests after their stay.
11	Guest_Folio	A summary table of guest expenses and costs during a booking period.
12	Folio_Item	Contains detailed line items (charges) associated with a specific folio.
13	Payment	Records detailed information regarding guest payment transactions.
14	Spa_Booking	Manages individual spa service appointments for guests.
15	Spa_Service	Catalog of spa services available for booking.
16	Treatment_Room	Manages the treatment rooms used for spa services.
17	Therapist	Store information and status of therapists when they are at work.
18	Schedule	Store schedule and slot to manage session of treatment room and therapist
19	Meal_Order	Manages food orders placed by guests.
20	Meal_Order_Item	Details the specific food items included in a meal order.
21	Menu_Item	Catalog of food items available on the menu.

2. Use Case Specifications - Đặc

*[Provide specifications for the use cases (UCs) those are covered in the system. The UCs are grouped by the system features and even sub features. **You just need to provide UC specifications for complex UCs involving in the main workflows (business processes).** Other UCs (i.e CRUD or data-viewing UCs) are simple, and you just need to refer the descriptions in the Functional Requirement (part 3) below]*

2.1 Authentication & Sensitive Health Profile

ID and Name:	UC-02 – Complete Health & Dietary Profile		
Primary Actor:	Guest	Secondary Actors:	System Administrator
Description:	This use case allows a guest to create and maintain a sensitive Health & Dietary Profile containing health conditions, food allergies, and dietary preferences. The		

	profile is used to personalize retreat experiences while ensuring privacy, consent management, and strict role-based access control.
Trigger:	Guest selects “Health & Dietary Profile” after successful authentication.
Preconditions:	- Guest account exists and is authenticated. - Guest account status is Active. - Guest has accepted Privacy Policy.
Postconditions:	- Health and dietary profile is encrypted and stored. - Authorized services may access only relevant information. - Audit logs are recorded.
Normal Flow:	1. Guest opens Health & Dietary Profile. 2. System displays profile form. 3. System displays consent statement with unchecked consent box by default (BR-08). 4. Guest enters dietary preferences. 5. Guest enters allergy information. 6. Guest enters health conditions. 7. Guest grants explicit consent. 8. Guest submits profile. 9. System validates required information. 10. System encrypts sensitive information before persistence (BR-09). 11. System stores profile. 12. System records audit log (BR-15). 13. System displays confirmation.
Alternative Flows:	• A1. Guest provides dietary profile only. → Health information remains empty. • A2. Guest updates existing profile. → System overwrites current values and records update history.
Exceptions:	• E1. Consent not granted. → System rejects submission and displays MSG-03. • E2. Unexpected storage failure. → System displays MSG-15.
Priority:	High
Frequency of Use:	Medium
Business Rules:	BR-07 – Role-Based Access Control and Data Minimization.

	BR-08 – Explicit consent is mandatory. BR-09 – Sensitive data encryption is required. BR-10 – Guest has the right to request permanent deletion. BR-15 – System audit logging is mandatory.
Other Information:	- Health information is classified as Sensitive Personal Data.
Assumptions:	- Guests provide accurate and updated information.

2.2 Retreat Package & Accommodation Booking

ID and Name:	UC07 – Book Retreat Package		
Primary Actor:	Guest	Secondary Actors:	Payment Gateway API
Description:	This use case allows a guest to browse retreat packages, select stay dates, choose a preferred Villa Type, and secure the reservation through a deposit payment.		
Trigger:	Guest selects “Book Retreat Package”.		
Preconditions:	- Guest is authenticated. - Retreat package exists and is active. - Villa inventory exists.		
Postconditions:	- Booking is created successfully. - Deposit transaction is stored. - Confirmation is generated. - Audit records are created.		
Normal Flow:	14. Guest opens Retreat Package page. 15. System displays available packages. 16. Guest selects package. 17. Guest selects stay dates. 18. Guest selects Villa Type only (BR-02). 19. System checks inventory. 20. System calculates package cost and deposit. 21. Guest confirms booking. 22. System redirects to Payment Gateway. 23. Guest completes deposit payment. 24. Gateway returns success result. 25. System confirms reservation (BR-01). 26. System generates itinerary.		

System Administration

	27. System records booking logs (BR-15). 28. System displays MSG-04.
Alternative Flows:	• A1. Guest changes package. → System recalculates pricing. • A2. Villa inventory changes. → System refreshes availability.
Exceptions:	• E1. Deposit payment failed. → System displays MSG-05. • E2. Booking confirmation timeout. → Reservation remains Pending.
Priority:	High
Frequency of Use:	High
Business Rules:	BR-01 – Booking requires successful deposit. BR-02 – Guest selects Villa Type only. BR-15 – Audit Trail is mandatory.
Other Information:	- Actual Villa assignment occurs during Check-in.
Assumptions:	- Payment services are available.

2.3 Spa & Therapy Scheduling Engine

ID and Name:	UC11 – Schedule Therapy Session		
Primary Actor:	Guest	Secondary Actors:	Spa Therapist / Yoga Trainer Calendar & Notification Service
Description:	This use case allows a guest with an active retreat booking to schedule therapy or wellness sessions included in the purchased retreat package. The system automatically coordinates therapist availability and treatment room availability to prevent resource conflicts.		
Trigger:	Guest selects “Schedule Therapy Session”.		
Preconditions:	- Guest has an active retreat booking.		

	- Guest has available therapy sessions included in the package. - Therapist schedule exists. - Treatment room availability exists.
Postconditions:	- Therapy session reservation is successfully created. - Therapist and treatment room allocation are completed. - Notification and reminder are generated. - Audit logs are stored.
Normal Flow:	29. Guest opens Therapy Scheduling. 30. System displays available therapy services. 31. Guest selects therapy type. 32. Guest selects preferred date and time. 33. System validates service eligibility based on purchased package (BR-05). 34. System checks therapist availability. 35. System checks treatment room availability. 36. System prevents resource collision and double booking (BR-04). 37. System allocates therapist and treatment room. 38. System creates therapy reservation. 39. System synchronizes reminder notification. 40. System records audit activity (BR-15). 41. System displays MSG-08.
Alternative Flows:	• A1. Guest purchases additional therapy service. → System allows reservation beyond package scope. • A2. Guest modifies session time. → System recalculates availability.
Exceptions:	• E1. No therapist or room available. → System displays MSG-09. • E2. Booking validation failed. → System rejects scheduling request.
Priority:	High
Frequency of Use:	High
Business Rules:	BR-04 – Two-dimensional Spa Scheduling. BR-05 – Spa service scope restriction. BR-15 – Audit Trail.
Other Information:	- Only assigned therapists are allowed to update session status.

Assumptions:	- Availability information is synchronized in real time.
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2.4 Dietary & Food Service Management.

ID and Name:	UC17 – View Daily Meal Preparation Dashboard		
Primary Actor:	F&B Staff / Chef	Secondary Actors:	None
Description:	This use case allows F&B staff to view meal requests and dietary constraints for guests in order to prepare meals that satisfy health restrictions and dietary requirements.		
Trigger:	Chef opens Daily Meal Dashboard.		
Preconditions:	- Guest meal selection exists. - Guest has active accommodation.		
Postconditions:	- Meal preparation dashboard is displayed. - Restricted information remains hidden.		
Normal Flow:	42. Chef opens Daily Meal Dashboard. 43. System retrieves meal orders. 44. System retrieves dietary preferences. 45. System retrieves allergy information only (BR-07). 46. System automatically filters incompatible menu items (BR-06). 47. System groups meal preparation requests. 48. System displays preparation instructions. 49. Chef confirms preparation readiness.		
Alternative Flows:	• A1. Guest updates meal preference. → Dashboard refreshes automatically. • A2. Menu becomes unavailable. → System recommends replacement dishes.		
Exceptions:	• E1. Unauthorized access attempt. → System rejects access and displays MSG-14. • E2. Missing dietary profile. → Dashboard shows warning.		
Priority:	Medium		
Frequency of Use:	High		

Business Rules:	BR-06 – Automatic Menu Filtering. BR-07 – RBAC and Data Minimization.
Other Information:	- Medical conditions are never shown to Chef.
Assumptions:	- Guest dietary profile exists before meal service.

2.5 Consolidated Billing & Checkout.

2.5.1 UC21 – Generate Consolidated Invoice

ID and Name:	UC21 – Generate Consolidated Invoice		
Primary Actor:	Receptionist	Secondary Actors:	Payment Gateway API
Description:	This use case allows the receptionist to consolidate all eligible charges into a final invoice before checkout.		
Trigger:	Receptionist initiates checkout.		
Preconditions:	- Guest booking exists. - Guest Folio exists. - Charges are posted.		
Postconditions:	- Consolidated invoice is generated. - Outstanding balance is calculated. - Audit logs are recorded.		
Normal Flow:	50. Receptionist opens Checkout. 51. System retrieves Guest Folio. 52. System retrieves Package charges. 53. System retrieves Spa charges. 54. System retrieves F&B charges. 55. System aggregates all charges using Room_Booking_ID (BR-11). 56. System deducts deposit amount (BR-12). 57. System calculates final payable amount. 58. System generates invoice. 59. System stores invoice record. 60. System records audit logs.		

Alternative Flows:	• A1. Additional services detected. → Invoice recalculation occurs. • A2. Promotion applies. → Final amount is adjusted.
Exceptions:	• E1. Transaction data missing. → System displays MSG-15.
Priority:	High
Frequency of Use:	High
Business Rules:	BR-11 – Guest Folio Consolidation. BR-12 – Checkout Constraint. BR-15 – Audit Trail.
Other Information:	- Only posted transactions appear in invoice.
Assumptions:	- All service usage has been synchronized.

2.5.2 UC22 – Process Final Payment

ID and Name:	UC22 – Process Final Payment		
Primary Actor:	Receptionist	Secondary Actors:	Payment Gateway API
Description:	This use case allows the receptionist to collect and process the final payment after all eligible charges have been consolidated into the final invoice. The payment process finalizes the guest stay and prepares checkout completion.		
Trigger:	Receptionist selects “Process Final Payment” after invoice confirmation.		
Preconditions:	- Consolidated invoice has been generated. - Guest booking status is Active. - Outstanding balance exists. - Payment service is available.		
Postconditions:	- Final payment is completed successfully.		

	- Invoice status becomes Paid. - Guest booking status becomes Completed. - Checkout becomes available. - Audit logs are stored.
Normal Flow:	61. Receptionist opens Final Payment screen. 62. System displays invoice summary. 63. System calculates remaining balance after deposit deduction (BR-12). 64. Receptionist confirms payment amount. 65. Guest selects payment method. 66. System redirects transaction to Payment Gateway. 67. Gateway validates payment. 68. System receives successful response. 69. System records transaction. 70. System updates invoice status to Paid. 71. System updates booking status to Completed. 72. System records audit logs (BR-15). 73. System displays payment success confirmation.
Alternative Flows:	• A1. Guest changes payment method. → System regenerates payment request. • A2. Payment requires additional verification. → System waits for asynchronous confirmation.
Exceptions:	• E1. Outstanding Spa/F&B fees still exist. → System rejects checkout and displays MSG-11. • E2. Payment failed. → Invoice remains unpaid. • E3. Gateway timeout occurs. → Transaction status becomes Pending. • E4. Unexpected payment exception. → System displays MSG-15.
Priority:	High
Frequency of Use:	High
Business Rules:	BR-12 – Checkout Constraint. BR-15 – Audit Trail.
Other Information:	- Checkout cannot be completed until payment is fully confirmed.

Assumptions:	- Payment gateway remains available.
--------------	--------------------------------------

2.6 Revenue Analytics & Reporting

2.6.1 UC24 – View Revenue Dashboard

ID and Name:	UC24 – View Revenue Dashboard		
Primary Actor:	Manager	Secondary Actors:	System Administrator
Description:	This use case allows managers to monitor business performance through analytical dashboards that summarize retreat revenue, occupancy indicators, therapist utilization, and food service performance.		
Trigger:	Manager selects “Revenue Dashboard”.		
Preconditions:	- Manager account is active. - Reporting data exists. - Data synchronization has completed.		
Postconditions:	- Dashboard information is displayed. - Reporting indicators become available for business decisions.		
Normal Flow:	74. Manager opens Revenue Dashboard. 75. System retrieves completed transaction records only (BR-13). 76. System retrieves retreat package revenue. 77. System retrieves Spa revenue. 78. System retrieves F&B revenue. 79. System calculates occupancy indicators. 80. System calculates therapist utilization. 81. System generates analytical charts. 82. Manager filters reporting period. 83. System refreshes dashboard. 84. Manager exports report if required.		
Alternative Flows:	• A1. Manager changes reporting dimensions. → Dashboard recalculates indicators. • A2. Manager exports dashboard. → System generates downloadable report.		
Exceptions:	• E1. Reporting data unavailable. → System displays empty dashboard. • E2. Aggregation calculation failed. → System displays MSG-15.		

Priority:	Medium
Frequency of Use:	Medium
Business Rules:	BR-13 – Reporting and Review Logic.
Other Information:	- Dashboard is read-only and cannot modify operational data.
Assumptions:	- Operational data synchronization runs successfully.

2.6.2 UC23 – Submit Retreat Review

ID and Name:	UC23 – Submit Retreat Review		
Primary Actor:	Guest	Secondary Actors:	None
Description:	This use case allows guests to submit ratings and reviews after completing their retreat experience.		
Trigger:	Guest selects “Submit Review”.		
Preconditions:	- Retreat booking status is Completed. - Checkout process has finished.		
Postconditions:	- Review is stored successfully. - Rating becomes available in reporting.		
Normal Flow:	85. Guest opens Review page. 86. System validates retreat completion status. 87. System displays review form. 88. Guest enters rating score. 89. Guest enters textual review. 90. Guest submits review. 91. System validates submission. 92. System stores review. 93. System displays successful confirmation.		
Alternative Flows:	• A1. Guest edits review. → System updates review.		

System Administration

Exceptions:	• E1. Guest has not completed retreat. → System rejects review submission. • E2. Unexpected system error. → System displays MSG-15.
Priority:	Low
Frequency of Use:	Medium
Business Rules:	BR-13 – Only completed stays may submit reviews.
Other Information:	- Reviews contribute to analytics and quality monitoring
Assumptions:	- Guests provide honest feedback.

3. Functional Requirements - Hải

3.1 Core Feature

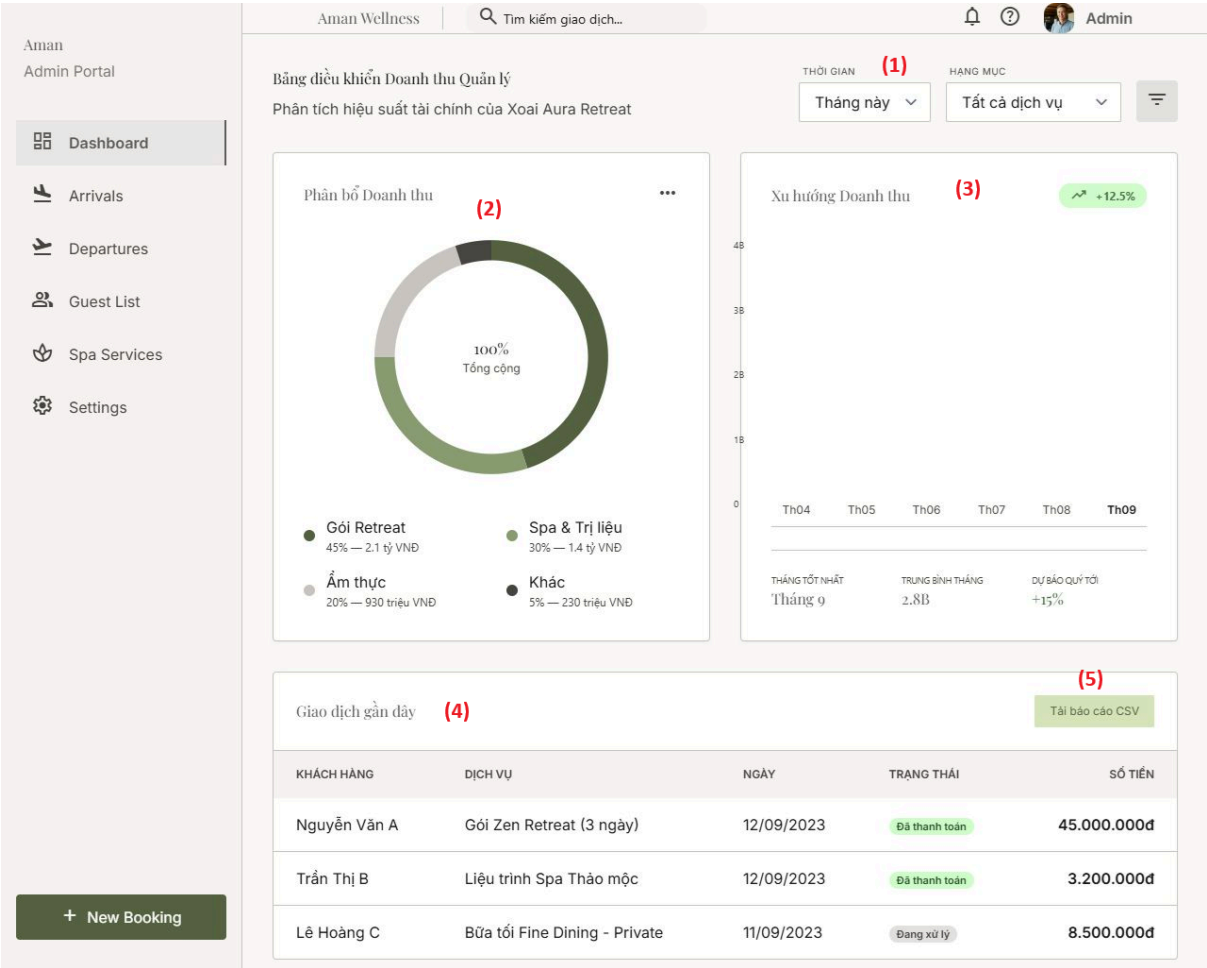
3.1.1 Health & Dietary Form Screen

[Content #1]

3.1.13 Manager Revenue Dashboard Screen

[Content #1]

- The screen layout utilizes a standard dashboard structure with a persistent left navigation sidebar.
- The main content area features a top filter bar, a mid-section containing two large data visualization widgets (a Donut chart and a Line/Bar chart), and a bottom section displaying a detailed data grid of financial transactions.



[Content #2]

- **Description:** This high-level dashboard provides the Management team with a comprehensive overview of the resort's financial performance. The system dynamically extracts and aggregates data from completed Consolidated Folios (Check-outs) to render real-time business intelligence metrics.
- **Mapped Use Case:** UC24 - As a Manager, I want to view a Revenue Dashboard (Pie/Bar charts) breaking down income by Retreat Packages, Spa, and F&B.

[Content #3]

Field Name	Description
Field Group: Dashboard Filters	
(1) Filters	Data type: Dropdown lists. - Time period: This Month, This Quarter, This Year. - Category: All Services, Retreat Packages, Spa, F&B. Action: Changing these values triggers a backend API call to re-query the financial data and dynamically re-render the associated chThe screen is divided into two main sections: "Diet & Allergies" on the left and "Physical Health Status" on the right. ● At the bottom of the form, there is a consent confirmation area and two action buttons.

Khai báo Sức khỏe & Ăn kiêng

Hồ sơ sức khỏe cá nhân của bạn giúp chúng tôi thiết kế hành trình chữa lành tối ưu. Mọi thông tin được bảo mật và chỉ sử dụng cho mục đích cá nhân hóa trải nghiệm.

(1)

Dị Ứng Thực phẩm

☐ Đậu phộng / Nuts

☐ Hải sản / Shellfish

☐ Sữa / Dairy

☐ Gluten

(2) Loại dị ứng khác...

(3) Chế độ Ăn kiêng

Thuần chay (Vegan)

Ăn chay (Vegetarian)

Keto

Halal

(4) Ghi chú bổ sung

Sở thích riêng hoặc yêu cầu đặc biệt về khẩu vị...

(5) Tình trạng Sức khỏe Vật Lý

Tình trạng Y tế Hiện tại

Ví dụ: Cao huyết áp, tiểu đường, hen suyễn...

(6) Thuốc đang sử dụng

Liệt kê các loại thuốc định kỳ nếu có...

(7) Chấn thương hoặc Vấn đề gần đây

Đau lưng, khớp, hoặc các vùng nhạy cảm cần lưu ý khi massage...

Sức khỏe của bạn là ưu tiên hàng đầu của Aman Wellness.

(8) ☐ Tôi xác nhận rằng thông tin cung cấp là chính xác và đồng ý để Aman Wellness sử dụng thông tin này nhằm mục đích cá nhân hóa trải nghiệm nghỉ dưỡng của tôi.

(9) LƯU & TIẾP TỤC

(10) XEM LẠI LỊCH TRÌNH

[Content #2]

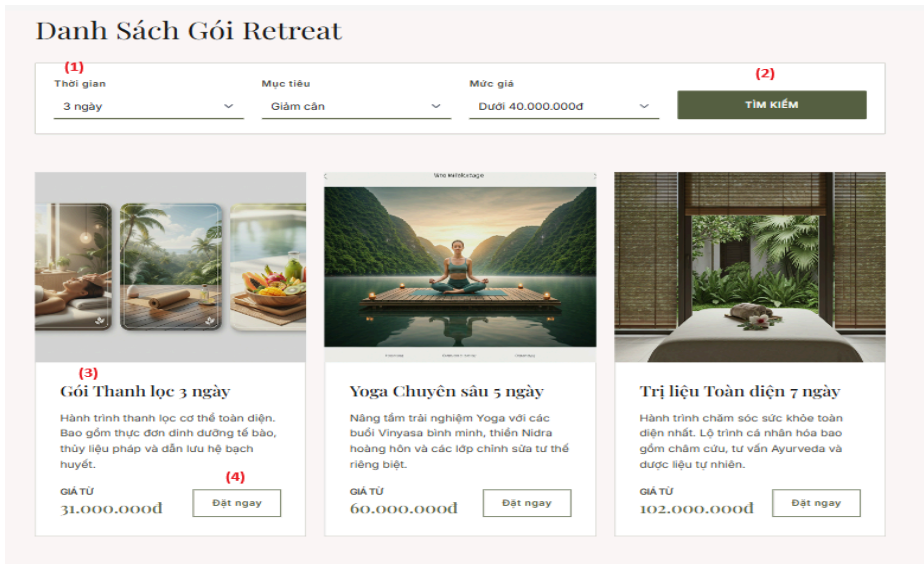
- **Description:** *This screen allows the Guest to input their personal health and dietary profile. This sensitive information is strictly confidential and will be segregated by the system: Chefs will only have access to dietary restrictions, while Spa Therapists will only view physical conditions.*
- **Mapped Use Case:** *UC02 - As a Guest, I want to complete my "Health & Dietary Profile".*

[Content #3]

Field Name	Description
Field Group: Dietary & Allergies	
(1) Food Allergies Checkboxes	Data type: Array of Booleans (Options: Peanuts, Shellfish, Dairy, Eggs, Soy, Wheat, Fish, and Seafood). Optional. Checked values will be routed to F&B Staff
(2) Other Allergies Input	Data type: String, max length of 255 characters. Option for listing other allergies not included in the predefined list.
(3) Diet Type Buttons	Data type: Enum/String (Options: Vegan, Vegetarian, Keto, Gluten-Free, and Halal). Required field for automatic menu filtering.

	(4) Additional Notes	Data type: String (Text area), max length of 500 characters. Specific taste preferences.
	Field Group: Physical Health Status	
	(5) Current Medical Conditions	Data type: String (Text area), max length of 500 characters. High blood pressure, diabetes).
	(6) Current Medications	Data type: String (Text area), max length of 500 characters.
	(7) Recent Injuries or Issues	Data type: String (Text area), max length of 500 characters. Back pain, joint issues). This specific data will be visible
	Field Group: Confirmation & Actions	
	(8) Consent Checkbox	Data type: Boolean. Strict Constraint: The initial value MUST be Unchecked with Personal Data Protection regulations (Decree 356/2018). The system must disable the Save button if this is not checked.
	(9) "Save & Continue" Button	Action: Triggers data submission. Backend must encrypt fields before inserting them into the Database. Redirect to dashboard.
	(10) "Review Itinerary" Button	Action: Cancels the current input and navigates back to dashboard.
	3.1.2 Data Erasure Request Screen [Content #1] - The UI is presented as a warning Modal/Dialog with prominent red text alerts. - It contains static warning text indicating that data cannot be recovered after deletion and a legal compliance note regarding personal data protection.	

	 Yêu cầu Xóa Dữ liệu Cá nhân Hành động này sẽ xóa vĩnh viễn toàn bộ hồ sơ sức khỏe, lịch trình hành trình và thông tin cá nhân của bạn khỏi hệ thống Xoai Aura. Dữ liệu không thể khôi phục sau khi xóa.  <i>Tuân thủ Nghị định 356/2025/NĐ-CP về bảo vệ dữ liệu cá nhân.</i> Nhập lại mật khẩu để xác nhận (1) (2)  (3) Hủy (4)  Xóa vĩnh viễn						
	[Content #2] • Description: This screen allows the Guest to exercise their "Right to Erasure" in compliance with data privacy regulations. The user is strictly required to re-authenticate via their password before the system executes the permanent deletion of their medical profiles, allergies, and itinerary records. • Mapped Use Case: UC05 - As a Guest, I want to exercise my "Right to Erasure", permanently deleting my health and allergy data from the system after the retreat ends. [Content #3] <table><tr><th>Field Name</th><th>Description</th></tr><tr><td colspan="2">Field Group: Verification</td></tr><tr><td>(1) Password Input</td><td>Data type: String. Required field. Input is masked by def to verify the user's identity before executing a destructi</td></tr></table>	Field Name	Description	Field Group: Verification		(1) Password Input	Data type: String. Required field. Input is masked by def to verify the user's identity before executing a destructi
Field Name	Description						
Field Group: Verification							
(1) Password Input	Data type: String. Required field. Input is masked by def to verify the user's identity before executing a destructi						

	(2) Toggle Visibility (Eye Icon)	Action: Toggles the input field (1) state between masked text (visible).
	Field Group: Actions	
	(3) "Hủy" (Cancel) Button	Action: Closes the warning dialog, aborts the operation to the previous screen.
	(4) "Xóa vĩnh viễn" (Permanent Delete) Button	Action: Submits the deletion request. - Validation: Validates the entered Password (1) against incorrect, display an inline error message. - Execution: If correct, execute a hard delete query to p sensitive health data from the database, terminate the log the user out.
3.1.3 Retreat Package List Screen [Content #1] - The UI features a horizontal search and multi-criteria filter bar (Duration, Goal, Price) fixed at the top of the section. - The main content area displays the available Retreat Packages in a responsive grid layout. Each package is contained within an individual card component featuring a thumbnail image, title, brief text description, starting price, and a call-to-action button.		
 The screenshot displays the 'Danh Sách Gói Retreat' (Retreat Package List) screen. At the top, there is a search bar with three filters: (1) 'Thời gian' (Duration) set to '3 ngày', (2) 'Mục tiêu' (Goal) set to 'Giảm cân' (Weight loss), and 'Mức giá' (Price) set to 'Dưới 40.000.000đ'. A 'TÌM KIẾM' (Search) button is on the right. Below the filters, there are three package cards. The first card, (3), is 'Gói Thanh lọc 3 ngày' (3-day Detox Package) with a price from 31.000.000đ and a (4) 'Đặt ngay' (Book now) button. The second card is 'Yoga Chuyên sâu 5 ngày' (5-day Deep Yoga Package) with a price from 60.000.000đ and a 'Đặt ngay' button. The third card is 'Trị liệu Toàn diện 7 ngày' (7-day Full-body Treatment Package) with a price from 102.000.000đ and a 'Đặt ngay' button. Each card features a thumbnail image and a brief description of the package.		

	[Content #2: Brief descriptions of the screen/function, mapped to the relevant use cases] • Description: This screen enables Guests to explore available Retreat Packages. Users can utilize multi-criteria filters to find itineraries that align with their personal wellness objectives. • Mapped Use Case: UC06 - As a Guest, I want to browse available "Retreat Packages" and filter them by goals (e.g., Weight Loss, Stress Relief, Yoga). [Content #3] <table border="1"> <thead> <tr> <th>Field Name</th><th>Description</th></tr> </thead> <tbody> <tr> <td colspan="2">Field Group: Verification</td></tr> <tr> <td>(1) Password Input</td><td>Data type: String. Required field. Input is masked by default to verify the user's identity before executing a destructive action.</td></tr> <tr> <td>(2) Toggle Visibility (Eye Icon)</td><td>Action: Toggles the input field (1) state between masked text (visible).</td></tr> <tr> <td colspan="2">Field Group: Actions</td></tr> <tr> <td>(3) "Hủy" (Cancel) Button</td><td>Action: Closes the warning dialog, aborts the operation to the previous screen.</td></tr> <tr> <td>(4) "Xóa vĩnh viễn" (Permanent Delete) Button</td><td> Action: Submits the deletion request. - Validation: Validates the entered Password (1) against incorrect, display an inline error message. - Execution: If correct, execute a hard delete query to permanently remove sensitive health data from the database, terminate the session and log the user out. </td></tr> </tbody> </table> 3.1.4 Package Detail & Checkout Screen [Content #1: UI Layout (Mockup screen prototype)] • The layout is divided into two main columns to optimize the user experience. • The left column displays the comprehensive details of the selected Retreat Package, including a large hero image, a descriptive text block outlining the itinerary, and highlighted included experiences. • The right column features a persistent booking form containing date pickers, a Villa type dropdown, guest information input fields, a	Field Name	Description	Field Group: Verification		(1) Password Input	Data type: String. Required field. Input is masked by default to verify the user's identity before executing a destructive action.	(2) Toggle Visibility (Eye Icon)	Action: Toggles the input field (1) state between masked text (visible).	Field Group: Actions		(3) "Hủy" (Cancel) Button	Action: Closes the warning dialog, aborts the operation to the previous screen.	(4) "Xóa vĩnh viễn" (Permanent Delete) Button	Action: Submits the deletion request. - Validation: Validates the entered Password (1) against incorrect, display an inline error message. - Execution: If correct, execute a hard delete query to permanently remove sensitive health data from the database, terminate the session and log the user out.
Field Name	Description														
Field Group: Verification															
(1) Password Input	Data type: String. Required field. Input is masked by default to verify the user's identity before executing a destructive action.														
(2) Toggle Visibility (Eye Icon)	Action: Toggles the input field (1) state between masked text (visible).														
Field Group: Actions															
(3) "Hủy" (Cancel) Button	Action: Closes the warning dialog, aborts the operation to the previous screen.														
(4) "Xóa vĩnh viễn" (Permanent Delete) Button	Action: Submits the deletion request. - Validation: Validates the entered Password (1) against incorrect, display an inline error message. - Execution: If correct, execute a hard delete query to permanently remove sensitive health data from the database, terminate the session and log the user out.														

dynamic cost summary block, and a prominent checkout button at the bottom.



Gói Thanh lọc & Tái tạo 5 Ngày

Hành trình tìm lại sự tĩnh tại và khôi phục năng lượng sống giữa lòng thiên nhiên hùng vĩ của Việt Nam. Xoai Aura Retreat mời bạn bước vào không gian của sự chữa lành, nơi thời gian dường như dừng lại.

Chương trình 5 ngày được thiết kế chuyên sâu bởi các chuyên gia sức khỏe, kết hợp giữa tinh hoa y học cổ truyền và liệu pháp hiện đại. Bạn sẽ được trải nghiệm chế độ dinh dưỡng hữu cơ "từ vườn đến bàn ăn", các buổi thiền định bên bờ suối và liệu pháp spa đặc quyền sử dụng thảo mộc bản địa.

Trải nghiệm bao gồm

 Lưu trú cao cấp
4 đêm tại Villa riêng biệt với tầm nhìn panorama.

 Ẩm thực thực dưỡng
Toàn bộ các bữa ăn hữu cơ theo chế độ riêng.

 Trị liệu Spa hàng ngày
60 phút massage và bấm huyệt mỗi ngày.

 Yoga & Thiền định
Các buổi tập riêng tư vào bình minh và hoàng hôn.

Thông tin Đặt phòng

(1)

NGÀY NHẬN PHÒNG

dd/mm/yyyy

NGÀY TRẢ PHÒNG

dd/mm/yyyy

LOẠI VILLA

(2)

Canopy Pavilion (Gần gũi thiên nhiên)

HỌ VÀ TÊN

(3)

Nguyễn Văn A

EMAIL

example@gmail.com

(4)

Giá gói (4 đêm)	42.500.000 VND
Thuế & Phí dịch vụ (10%)	4.250.000 VND
Tổng cộng	46.750.000 VND
Đặt cọc (30%)	14.025.000 VND

Thanh toán Đặt cọc

(5)

Thanh toán

Bảng cách nhân nút, bạn đồng ý với các [Điều khoản Dịch vụ](#) và [Chính sách Hoàn hủy](#) của Xoai Aura.

[Content #2]

- **Description:** This screen displays in-depth details of a selected package. Here, the Guest finalizes their itinerary dates, selects their accommodation, and securely pays the deposit.
- **Mapped Use Case:** UC07 - As a Guest, I want to select a package, pick travel dates, choose a Villa type, and securely pay the deposit.

[Content #3]

Field Name	Description
Field Group: Booking Form	
(1) Check-in/Check-out Dates	Data type: <code>LocalDate</code> .

		Constraint: Past dates are disabled. Check-out date must be greater than Check-in date based on the package's predefined duration.
	(2) Loại Villa (Villa Type)	Data type: Dropdown list. Constraint: The system must dynamically filter this list to only show villa types that have actual availability during the selected date range.
	(3) Guest Info (Name & Email)	Data type: String. Should be auto-populated if the Guest is already authenticated via SSO/Login.
	Field Group: Payment & Checkout	
	(4) Price Summary	Read-only calculation block. Dynamically updates based on selected Package Type. Displays Base Price, Tax (10%), Total, and the required deposit amount.
	(5) "Thanh toán" (Pay Deposit) Button	Action: - Integrates with and triggers the external Payment Gateway (Stripe / VNPay Sandbox) to handle secure transactions. - On Success: Generates a new Room_Booking_ID and sets the booking status.
3.1.5 Arrivals & Check-in Dashboard Screen [Content #1] - The layout is divided into a fixed left navigation sidebar and a primary main content area on the right. - The main content area features a date header, a detailed Data Table listing all expected arrivals for the current date, and summary metric cards at the bottom.		

Arrivals (1)

Departures

Guest List

Room Management

New Reservation

Help Center

Log Out

Danh sách khách đến hôm nay

14 Tháng 10, 2023 • 12 Guests Expected

TÊN KHÁCH HÀNG	GÓI RETREAT	VILLA	SỐ PHÒNG	TRẠNG THÁI	THAO TÁC
NH Nguyễn Văn Hoàng ID: #AUR-9231	Deep Meditation 7 Ngày	Forest View Villa	101	Chưa đến	Check-in
LT Lê Thị Thu Thủy ID: #AUR-9240	Detox & Yoga	Ocean Suite	206	Đang chờ	Check-in
TM Trần Minh Quân ID: #AUR-9245	Silent Retreat 3 Ngày	Garden Zen Villa	Chọn	Chưa đến	Check-in
PA Phạm Anh Thư ID: #AUR-9228	Aura Signature Spa	Hilltop Villa	402	Đã Check-in	Hoàn tất

CHỜ CHECK-IN

08 Khách

ĐÃ CHECK-IN

04 Khách

PHÒNG SẴN SANG

15 Villas

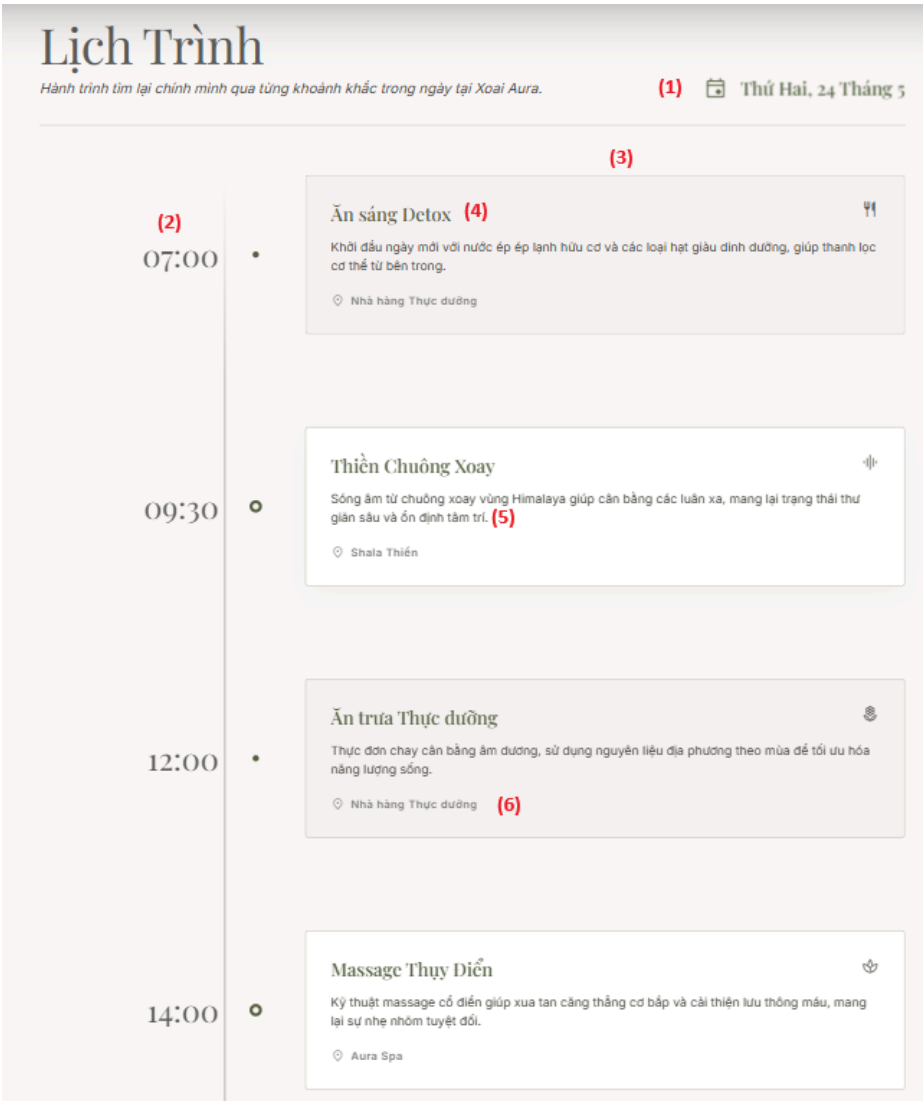
[Content #2]

- Description:** This is the primary operational dashboard for Receptionists. The system automatically fetches and displays all reservations scheduled for arrival on the current day. Receptionists use this interface to assign physical room numbers and process the check-in workflow.
- Mapped Use Case:** UC08 - As a Receptionist, I want to view a dashboard of expected arrivals and perform Check-In (assign a specific Villa room number).

[Content #3]

	Field Name	Description
	Field Group: Dashboard Overview	
	(1) Sidebar Navigation	Navigation links for front-desk operations (e.g., Arrivals, Management).
	(6) Summary Cards	Read-only numeric data. Dynamically counts and displays Pending Check-ins, Completed Check-ins, and Ready Villa
	Field Group: Arrivals Data Table	
	(2) Guest & Package Info	Data type: Text. Displays Guest Name, Booking ID, Retrieved booked Villa Category. Strict RBAC Constraint: For the "Special Requests" column query MUST mask/hide any Health or Dietary Allergy data for Receptionist role. Only general service requests (e.g., Water pillow) are permitted to be shown.
	(3) Số phòng (Room Assignment)	Data type: Dropdown List. Constraint: Upon clicking, the system must dynamically filter with specific room numbers that meet two conditions so the room matches the booked Villa Category AND its physical status is "Vacant/Ready".
	(4) Trạng thái (Status Badge)	Visual indicator of the booking status (e.g., Not Arrived, Pending).
	(5) "Check-in" Action Button	Action: Triggers the check-in execution. - Validation: The Receptionist must select a Room Number before submission. - Execution: Updates the Booking status to "Checked-in" and updates the physical Villa status (UC09) to "Occupied" if available.
	3.1.6 Guest Itinerary Timeline Screen	
	[Content #1]	
	The UI is structured as a vertical Timeline layout.	

- The top section contains the header and date navigation. The main body is split into a fixed time axis on the left and corresponding event detail cards on the right.



[Content #2]

- **Description:** This screen provides Guests with a visual and detailed overview of all scheduled daily activities at the resort. It seamlessly aggregates data from various sub-systems (such as F&B meal plans, Yoga classes, and Spa therapy sessions) into a single, continuous chronological flow.
- **Mapped Use Case:** UC10 - As a Guest, I want to view my full booking details and itinerary timeline.

[Content #3]

	Field Name	Description
	Field Group: Date Navigation	
	(1) Date Selector / Display	Data type: LocalDate. Action: Displays the current itinerary date (e.g., Monday) and allows users to click and navigate between different days within their booked Retreat Package.
	Field Group: Timeline Events	
	(2) Time Milestone	Data type: LocalTime (e.g., 07:00, 09:30). Displays the scheduled event.
	(3) Activity Card	Read-only container. Dynamically queries and aggregates data from the Spa scheduling (Module 3) and F&B planning (Module 4).
	(4) Activity Title & Icon	Data type: String & Icon Asset. Displays the title of the event (e.g., "Ăn sáng Detox", "Massage thư giãn") accompanied by a context-specific icon in the top right corner.
	(5) Activity Description	Data type: String. A brief descriptive text outlining the various benefits.
	(6) Location	Data type: String. Displays the physical venue retrieved from the master data (e.g., "Thực dưỡng", "Shala Thiền", "Aura Spa").
	3.1.7 Guest Spa Scheduler Screen [Content #1] - The layout is organized into two distinct columns. The left column displays the selected service details and an interactive monthly Calendar component. - The right column features a responsive grid of available time slots and a fixed Booking Summary block at the bottom, containing the primary call-to-action button.	

Đặt lịch Trị liệu Spa

Trải nghiệm sự tinh lặng và tái tạo năng lượng trong không gian thiền định của Xoai Aura.

Dịch vụ đã chọn

Massage Thụy Điển - 60 phút

2.400.000 VND

1. CHỌN NGÀY

Tháng 12, 2024

T2	T3	T4	T5	T6	T7	CN
25	26	27	28	29	30	1
2	3	4	5	6	7	8
9	10	11	12	13	14	15
16	17	18	19	20	21	22
23	24	25	26	27	28	29
30	31					

2. CHỌN GIỜ

09:00
Sáng

10:30
Sáng

14:00
Chiều

16:00
Chiều

17:30
Chiều

19:00
Hết chỗ

TÓM TẮT ĐẶT LỊCH

Ngày:

11 Tháng 12, 2024

Thời gian:

10:30 (60 phút)

Phòng:

Lotus Suite 02

Tổng cộng:

2.400.000 VND

Xác nhận đặt lịch

Hủy lịch miễn phí trước 24 giờ

[Content #2]

- **Description:** This interface empowers Guests to schedule their included Spa therapies. Crucially, the system acts as a background orchestrator, computing real-time resource availability (rooms and staff) to dynamically render valid time slots, thereby completely preventing overbooking scenarios.
- **Mapped Use Case:**
 - UC11 - As a Guest, I want to schedule included Spa sessions by selecting a date and time slot.
 - UC12 - As a System, I must automatically find an available slot by simultaneously matching ONE free Therapist AND ONE empty Treatment Room.

[Content #3]

Field Name	Description
Field Group: Scheduling Inputs	

	(1) Service Info	Read-only container. Displays the Service Name, Duration, Price.
	(2) Date Picker	Data type: LocalDate . Constraint: Past dates are strictly disabled. Selecting a valid date triggers an API request to the backend to fetch the corresponding available Time Slot Grid (3).
	(3) Time Slot Grid	Data type: LocalTime . Strict Business Rule: A time slot button is "Clickable/Available" IF the Backend query verifies the availability of (Count of Therapist_ID > 0) for that exact duration. Otherwise, it is rendered as Disabled/Grayed-out (e.g., "19:00 Hết chỗ"). This implements 2-Dimensional Double-Booking Prevention.
	Field Group: Confirmation	
	(4) Booking Summary	Read-only block. Dynamically updates based on inputs from the previous fields. Logic Note: The "Room" field (e.g., Lotus Suite 02) is automatically assigned by the system's matching algorithm (UC12); the guest does not select a room.
	(5) "Xác nhận đặt lịch" (Confirm Booking) Button	Action: Submits the booking request. Backend Constraint: The execution must be enclosed in a Database Transaction. The system must simultaneously allocate the selected Therapist and Room resources to strictly prevent concurrency issues (Double-booking) during the write operation.
	(5) Activity Description	Data type: String. A brief descriptive text outlining the service and its benefits.
3.1.8 Therapist Daily Schedule Screen [Content #1] The screen layout consists of a Header displaying the current operational date (e.g., Tuesday, June 2, 2026) alongside secondary action buttons.		

- Below the header are three prominent summary KPI cards.
- The core element is a chronological Data Table detailing the therapist's assigned treatment sessions for the day.



[Content #2]

- **Description:** This specialized dashboard empowers Spa Therapists to manage their daily shifts. It provides a structured view of upcoming appointments, grants secure access to permitted physical health records for treatment preparation, and allows therapists to update real-time session statuses.
- **Mapped Use Case:**
 - UC13 - As a Spa Therapist, I want to view my daily schedule and access specific medical notes of assigned guests.
 - UC14 - As a Spa Therapist, I want to mark a session as "Completed" or "No-Show".

[Content #3]

Field Name	Description
Field Group: Dashboard Summary	
(1) Date Header	Data type: <code>LocalDate</code> . Defaults to the current system

	(2) Summary Cards	Read-only dynamic metrics (Total sessions, Completed sessions, currently Vacant rooms) calculated from the active data
	Field Group: Schedule Data Table	
	(3) Session Details	Read-only text. Displays Time Range, Guest Name, Treatment, Assigned Room. This data is the direct output of the algorithm (UC12).
	(4) "Xem Ghi Chú" (View Notes) Button	Action: Triggers a Modal dialog displaying the guest's notes. Strict RBAC Constraint: The Backend API serving this view MUST implement Data Minimization. It is strictly restricted to non-physical treatment-related data (e.g., injuries, back pain). It must completely mask/exclude any dietary allergy information.
	(5) Trạng thái (Status Dropdown)	Data type: Enum. Allowed values: Đang chờ (Pending), Hoàn thành (Completed), Hủy (Cancelled), Không xuất hiện (No-Show). Action: Updating this field commits the state change. Marking a session as "Completed" triggers the billing process. Marking as "Cancelled" or "No-Show" triggers the guest's Consolidated Folio (Module 5).
3.1.9 Manual Spa Booking Modal [Content #1] - The interface is rendered as an overlay dialog (Modal) to maintain the user's current context, featuring a close (X) icon at the top right. - It contains sequential input fields for guest identification, service selection, scheduling, and a dynamic price summary block.		

Đặt dịch vụ Spa bổ sung

Cập nhật dịch vụ vào Folio khách hàng

TÊN KHÁCH HÀNG HOẶC SỐ PHÒNG (1)

Tim kiếm khách đang lưu trú...

DỊCH VỤ SPA (2)

Chọn dịch vụ...

NGÀY SỬ DỤNG (3)

24/05/2024

KHUNG GIỜ CÒN TRỐNG (4)

09:00	10:30	14:00	15:30
17:00	18:30	20:00	

(5)

Trị liệu da nóng (90p)	1,850,000 đ
Phí dịch vụ & Thuế (15%)	277,500 đ
TỔNG CỘNG	2,127,500 đ

(6)

Xác nhận & Ghi nợ vào Folio

Hủy

[Content #2]

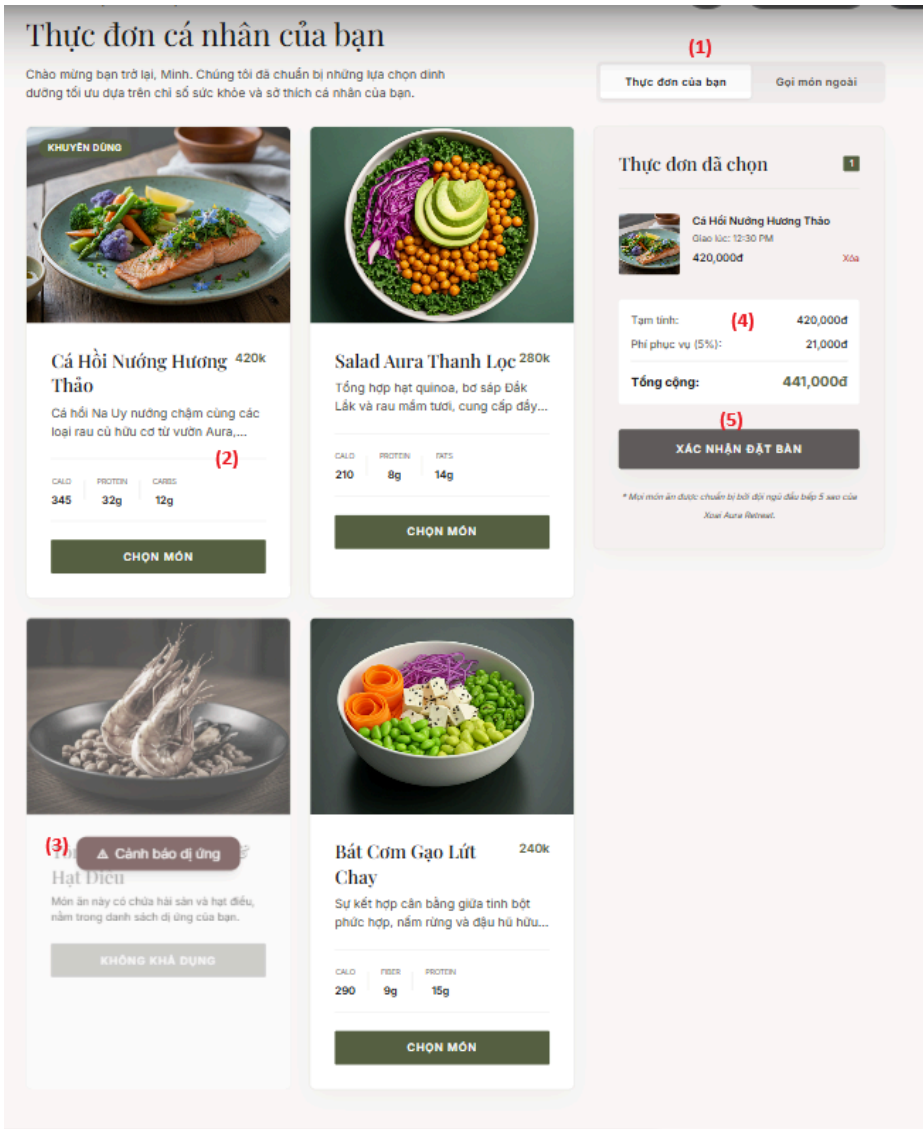
- **Description:** This operational function enables Receptionists to manually book additional, a-la-carte Spa therapies for in-house guests. Utilizing standard hospitality accounting mechanics, the system bypasses immediate point-of-sale payment and routes the charges directly to the guest's centralized room account.
- **Mapped Use Case:** UC15 - As a Receptionist, I want to manually book additional Spa services for guests and charge them to their Villa folio.

[Content #3]

Field Name	Description
------------	-------------

	Field Group: Guest & Service Selection	
	(1) Guest/Room Search Input	Data type: String (Autocomplete). Constraint: The backend query must filter and return only current reservation status is actively "Checked-in".
	(2) Spa Service	Data type: Dropdown List. Populated from the Spa Master List.
	Field Group: Scheduling	
	(3) Date	Data type: LocalDate . Defaults to the current system date.
	(4) Available Time Slots	Data type: LocalTime grid. Logic: Reuses the backend validation algorithm to determine available slots where both a Treatment Room and a Therapist are available.
	Field Group: Folio Integration	
	(5) Price Summary	Read-only calculation block. Displays Base Price, Taxes/Service Fees, and the Grand Total.
	(6) "Xác nhận & Ghi nợ vào Folio" (Confirm & Post to Folio) Button	Action: Form submission. Strict Backend Constraint: This action must execute a dual-operation database transaction: 1. Insert a new Spa appointment record. 2. Post the financial charge to the central Guest Folio using Room_Booking_ID as the primary foreign key link. The reservation status remains "Pending" until the Consolidated Billing process is complete at Check-out (UC21).
	3.1.10 Personalized Menu & A-la-carte Screen [Content #1] The layout is split into a main catalog area on the left and a persistent floating Cart/Summary block on the right.	

- The catalog area features top navigation tabs to switch between included package meals and extra a-la-carte options. Food items are displayed as detailed cards containing nutritional metrics.



[Content #2]

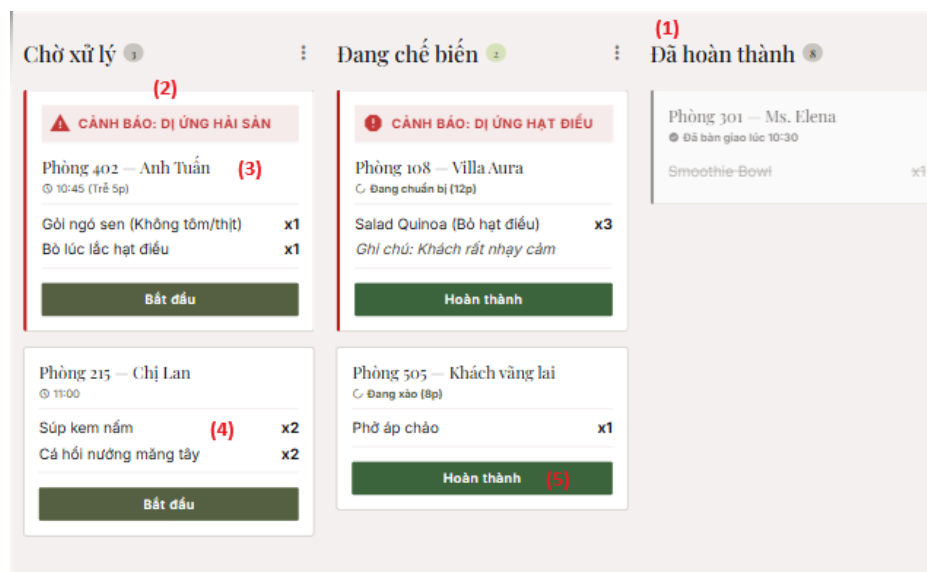
- **Description:** This interface delivers a smart dining experience. The system automatically cross-references dish ingredients against the user's health profile to issue warnings and prevent the ordering of allergens. Furthermore, guests can seamlessly order premium items outside their standard package via the a-la-carte tab.
- **Mapped Use Case:**

	○ UC16 - As a Guest, I want to pre-select my daily meals from a menu automatically filtered by the system based on my allergy/dietary profile. ○ UC19 - As a Guest, I want to order extra a-la-carte food/drinks outside my package and charge it to the Villa. [Content #3] <table border="1"> <thead> <tr> <th>Field Name</th><th>Description</th></tr> </thead> <tbody> <tr> <td colspan="2">Field Group: Menu Navigation</td></tr> <tr> <td>(1) Category Tabs</td><td> Action: Toggles the active dataset. - Thực đơn của bạn (Your Menu): Renders items in Package (Zero cost). - Gọi món ngoài (A-la-carte): Renders the premium associated monetary values. </td></tr> <tr> <td colspan="2">Field Group: Food Items</td></tr> <tr> <td>(2) Standard Food Card</td><td>Read-only elements: Image, Dish Name, Ingredients De Facts (Calories, Macros), and an active "Chọn món" (Ad</td></tr> <tr> <td>(3) Allergy Warning Card</td><td> Strict Logic Constraint: During data retrieval, the perform an intersection check between the dish's I and the guest's Food_Allergies array. If a match Frontend must render an alert overlay (e.g., "Cảnh strictly disable the selection button to prevent the it added to the cart. </td></tr> <tr> <td colspan="2">Field Group: Cart & Checkout</td></tr> <tr> <td>(4) Cart Summary</td><td>Dynamic calculation block. Orders from the include 0 VND. Orders from the A-la-carte tab dynamically + Service Fees.</td></tr> <tr> <td>(5) "Xác nhận đặt bàn" (Confirm Order) Button</td><td> Action: Submits the Order payload to the database Folio Constraint: For any items sourced from the system must automatically post the calculated final to the guest's centralized Folio account associated Room_Booking_ID </td></tr> </tbody> </table>	Field Name	Description	Field Group: Menu Navigation		(1) Category Tabs	Action: Toggles the active dataset. - Thực đơn của bạn (Your Menu): Renders items in Package (Zero cost). - Gọi món ngoài (A-la-carte): Renders the premium associated monetary values.	Field Group: Food Items		(2) Standard Food Card	Read-only elements: Image, Dish Name, Ingredients De Facts (Calories, Macros), and an active "Chọn món" (Ad	(3) Allergy Warning Card	Strict Logic Constraint: During data retrieval, the perform an intersection check between the dish's I and the guest's Food_Allergies array. If a match Frontend must render an alert overlay (e.g., "Cảnh strictly disable the selection button to prevent the it added to the cart.	Field Group: Cart & Checkout		(4) Cart Summary	Dynamic calculation block. Orders from the include 0 VND. Orders from the A-la-carte tab dynamically + Service Fees.	(5) "Xác nhận đặt bàn" (Confirm Order) Button	Action: Submits the Order payload to the database Folio Constraint: For any items sourced from the system must automatically post the calculated final to the guest's centralized Folio account associated Room_Booking_ID
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3.1.11 Chef Dashboard (Kitchen Display System)

[Content #1]

- The interface employs a Kanban board layout, segmented into three primary operational columns: Pending, In Progress, and Completed.
- Each culinary order is represented as a dynamic Ticket/Card. These cards display room assignments, wait timers, itemized dish lists, and quick-action buttons. Critical allergy alerts are prominently anchored at the top of applicable cards in high-contrast red.

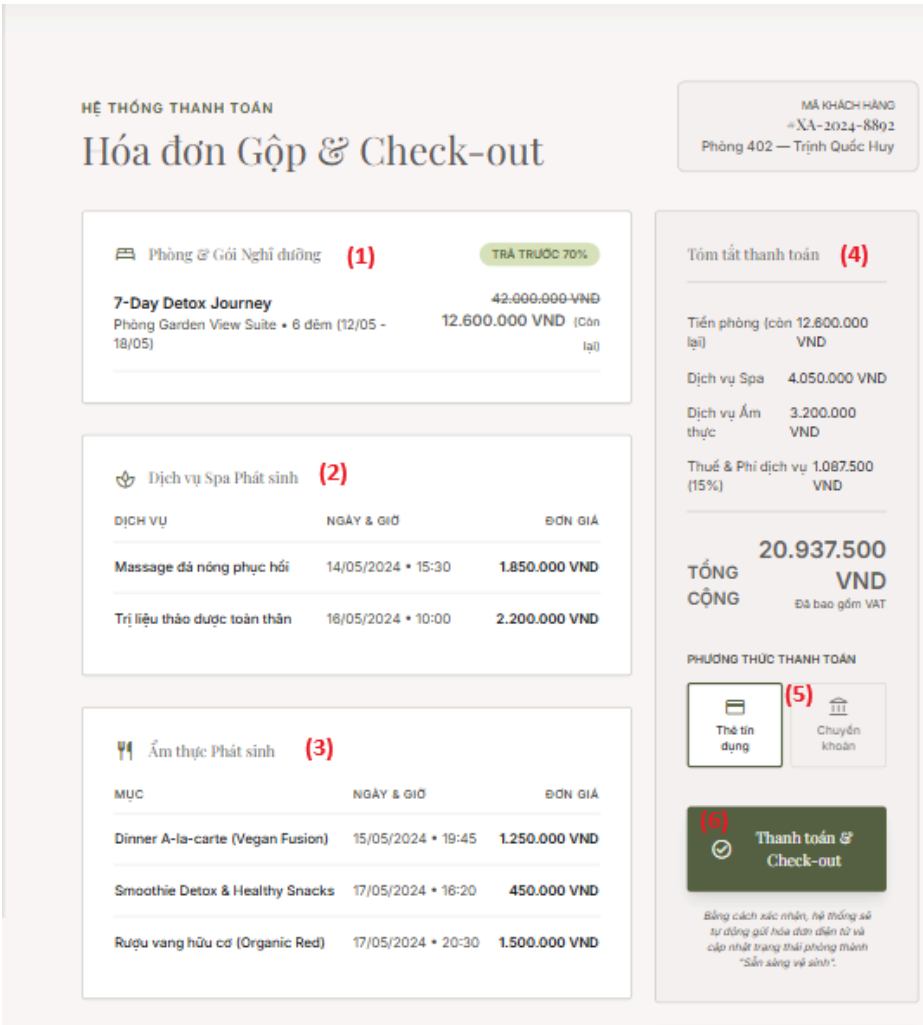


[Content #2]

- **Description:** This real-time operational dashboard is utilized by Chefs and F&B staff. It digitizes the ticket routing process, enabling the kitchen to monitor preparation workflows and strictly adhere to food safety constraints without violating patient medical privacy.
- **Mapped Use Case:**
 - UC17 - As a Chef, I want to view a "Daily Meal Prep Dashboard" aggregating orders and specific allergy alerts.
 - UC18 - As a Chef, I want to update the status of an order ticket (e.g., Preparing -> Ready).

	○ UC20 - As a System, I must mask the Guest's entire medical history from the Chef, displaying ONLY relevant "Food Allergies". [Content #3] <table border="1"> <thead> <tr> <th>Field Name</th><th>Description</th></tr> </thead> <tbody> <tr> <td colspan="2">Field Group: Kanban Columns</td></tr> <tr> <td>(1) Order Status Columns</td><td>Layout containers that group tickets by their current status: In_Progress, and Completed</td></tr> <tr> <td colspan="2">Field Group: Order Ticket Details</td></tr> <tr> <td>(2) Allergy Alert Label</td><td> Data type: String/Alert Banner. Strict RBAC / Data Minimization Constraint: The Backend of this UI MUST restrict the payload to "Dietary Allergies" (e.g., seafood). Physical medical records (e.g., back pain, hypertension) are completely masked and excluded from this view. </td></tr> <tr> <td>(3) Order Meta Info</td><td>Read-only text. Displays Room Number, Guest Name, and wait Timer tracking ticket aging (e.g., "Trễ 5p")</td></tr> <tr> <td>(4) Itemized List</td><td>Data type: Array of Objects. Displays dish names, prices, and specific dietary modifications/notes associated with each item.</td></tr> <tr> <td>(5) State Mutation Buttons</td><td> Action Buttons ("Bắt đầu" / Start and "Hoàn thành" / Complete) Action: Triggers a state update query in the database, migrating the ticket to the next logical Kanban column. </td></tr> </tbody> </table> 3.1.12 Consolidated Billing & Check-out Screen [Content #1] • The layout utilizes a 7:3 two-column structure. The wider left column provides a granular breakdown of all incurred charges, categorized logically by operational departments (Room, Spa, F&B).	Field Name	Description	Field Group: Kanban Columns		(1) Order Status Columns	Layout containers that group tickets by their current status: In_Progress , and Completed	Field Group: Order Ticket Details		(2) Allergy Alert Label	Data type: String/Alert Banner. Strict RBAC / Data Minimization Constraint: The Backend of this UI MUST restrict the payload to "Dietary Allergies" (e.g., seafood). Physical medical records (e.g., back pain, hypertension) are completely masked and excluded from this view.	(3) Order Meta Info	Read-only text. Displays Room Number, Guest Name, and wait Timer tracking ticket aging (e.g., "Trễ 5p")	(4) Itemized List	Data type: Array of Objects. Displays dish names, prices, and specific dietary modifications/notes associated with each item.	(5) State Mutation Buttons	Action Buttons ("Bắt đầu" / Start and "Hoàn thành" / Complete) Action: Triggers a state update query in the database, migrating the ticket to the next logical Kanban column.
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- The narrower right column is a sticky sidebar displaying the grand total summary, payment method selectors, and the primary check-out execution button.



[Content #2]

- **Description:** This screen digitizes the hotel night audit and central accounting processes. Functioning as a centralized Guest Folio, the system automatically aggregates all pending financial liabilities from various Point-of-Sale (POS) terminals across the resort into a single, comprehensive final invoice.
- **Mapped Use Case:**
 - UC21 - As a Receptionist, during Check-Out, I want to generate a Consolidated Bill summarizing the remaining Package Cost, extra Spa services, and extra F&B orders.

	○ UC22 - As a Receptionist, I want to process the final payment and change the Villa status to Vacant/Needs Cleaning.															
	[Content #3]															
	<table><tr><th>Field Name</th><th>Description</th></tr><tr><td colspan="2">Field Group: Consolidated Folio Details</td></tr><tr><td>(1), (2), (3) Departmental Charge Blocks</td><td>Read-only Data Grids. Backend Logic: The system must automatically aggregate and execute cross-module queries (Modules 2, 3, and 4), using Room_Booking_ID as the primary relational linkage.</td></tr><tr><td colspan="2">Field Group: Payment Processing</td></tr><tr><td>(4) Payment Summary</td><td>Read-only calculation block. Displays the algebraic sum of Room balance, Spa charges, F&B charges, and applicable taxes.</td></tr><tr><td>(5) Payment Method</td><td>Data type: Toggle/Radio selection (e.g., Credit Card, Debit Card, Cash).</td></tr><tr><td>(6) "Thanh toán & Check-out" (Pay & Check-out) Button</td><td>Action: Executes the final transaction and terminates the current stay. Strict Business Constraint: The system must validate that all associated sub-orders are completed. Guests CANNOT check-out with pending/unprocessed Spa or F&B orders (the button is disabled/throw an error). Post-execution: Upon successful payment, the system updates the booking status to "Checked-out" and automatically changes the physical Villa status to Vacant/Needs Cleaning.</td></tr></table>		Field Name	Description	Field Group: Consolidated Folio Details		(1), (2), (3) Departmental Charge Blocks	Read-only Data Grids. Backend Logic: The system must automatically aggregate and execute cross-module queries (Modules 2, 3, and 4), using Room_Booking_ID as the primary relational linkage.	Field Group: Payment Processing		(4) Payment Summary	Read-only calculation block. Displays the algebraic sum of Room balance, Spa charges, F&B charges, and applicable taxes.	(5) Payment Method	Data type: Toggle/Radio selection (e.g., Credit Card, Debit Card, Cash).	(6) "Thanh toán & Check-out" (Pay & Check-out) Button	Action: Executes the final transaction and terminates the current stay. Strict Business Constraint: The system must validate that all associated sub-orders are completed. Guests CANNOT check-out with pending/unprocessed Spa or F&B orders (the button is disabled/throw an error). Post-execution: Upon successful payment, the system updates the booking status to "Checked-out" and automatically changes the physical Villa status to Vacant/Needs Cleaning.
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arts.																
Field Group: Data Visualizations																
(2) Revenue Breakdown	Display type: Donut / Pie Chart widget. Display Logic: Visually represents the percentage (%) and absolute monetary value segmented by the primary revenue streams: Retreat Packages, Spa & Therapy, and F&B.															
(3) Revenue Trend	Display type: Line / Bar Chart widget.															

	Display Logic: Illustrates the fluctuation of total revenue across a timeline (e.g., months). It also includes static KPI summaries such as Best Month and Monthly Average.
Field Group: Transaction Data	
(4) Recent Transactions Table	Read-only Data Grid. Displays a paginated list of the most recent completed check-out transactions, detailing Guest Name, Service, Date, Status, and Amount.
(5) "Tải báo cáo CSV" (Export CSV) Button	Action: Triggers a backend service to export the dataset from table (4), based on the active filters, and downloads it to the user's device in <code>.csv</code> or <code>.xlsx</code> format.

3.2 User Authentication

3.2.1 Authentication & Login Screen

[Content #1]

- The user interface features a clean, centrally-aligned authentication card utilizing elegant serif typography.
- It accommodates a traditional credential-based input form, visually separated by an "OR" divider, followed by a seamless Single Sign-On (SSO) option via Google. A directional link for new user registration is anchored at the bottom.

Chào mừng trở lại

Đăng nhập để tiếp tục hành trình tinh thức của bạn.

EMAIL (1)

email@example.com

MẬT KHẨU (2)

Quên mật khẩu? (3)

(4)

Đăng nhập →

HOẶC

(5)

 Google

(6)

Chưa có tài khoản? Đăng ký ngay

[Content #2]

- **Description:** This screen serves as the primary security gateway to the application. It securely validates user identities and establishes active sessions. Users are provided the flexibility to authenticate via standard credentials or leverage Google's OAuth2 service for expedited access.
- **Mapped Use Case:** UC01 - Log in to the system (Includes traditional and SSO authentication).

[Content #3]

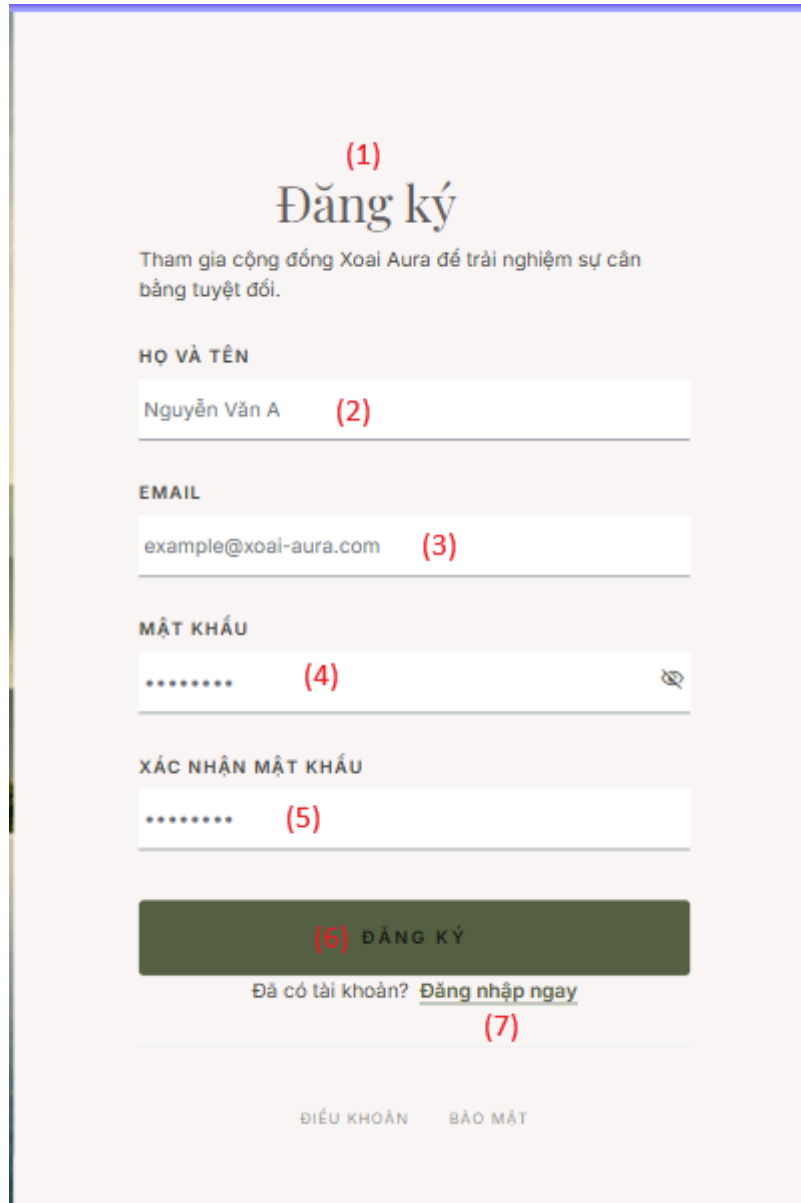
Field Name	Description
Field Group: Traditional Login	
(1) EMAIL	Data type: String. Required field with strict email format validation.
(2) MẬT KHẨU (Password)	Data type: String. Required field. User input is securely masked by default.
(3) "Quên mật khẩu?" (Forgot Password) Link	Action: Navigates the user to the Password Recovery workflow interface (UC04).
(4) "Đăng nhập" (Login) Button	Action: Submits the credential payload to the authentication endpoint. Security Constraint: The backend must utilize secure hashing algorithms (e.g., BCrypt) for password verification. Upon successful

	authentication, the server provisions a valid JSON Web Token to the client.
Field Group: Alternative Authentication	
(5) "Google" SSO Button	Action: Initializes the external OAuth2 authentication flow with Google. Logic: Bypasses manual password entry. The backend securely decodes the received Google token to extract the unique Email identifier and seamlessly authorize system access
(6) "Đăng ký ngay" (Register Now) Link	Action: Directs unauthenticated users to the New Account Registration screen (UC03).

3.2.2 New Account Registration Screen

[Content #1]

- The registration form interface is minimalist, maintaining visual and structural consistency with the Authentication screen.
- Input fields employ a clean, stacked layout with borderless underlines. The primary call-to-action button is prominently displayed, followed by secondary navigation links at the footer.



The image shows a minimalist registration form for 'Xoai Aura'. It features a light pink background and a white form area. The form is titled 'Đăng ký' (Register) in a large, elegant font, with a red '(1)' above it. Below the title is a short description in Vietnamese. The form consists of several input fields: 'HỌ VÀ TÊN' (Full Name) with a red '(2)' above it, 'EMAIL' with a red '(3)' above it, 'MẬT KHẨU' (Password) with a red '(4)' above it, and 'XÁC NHẬN MẬT KHẨU' (Confirm Password) with a red '(5)' above it. A prominent dark green button with the text 'ĐĂNG KÝ' (Register) and a red '(6)' above it is located below the password fields. Below the button is a link 'Đã có tài khoản? Đăng nhập ngay' (Already have an account? Log in now) with a red '(7)' below it. At the bottom of the form are two links: 'ĐIỀU KHOẢN' (Terms) and 'BẢO MẬT' (Privacy Policy).

(1) Đăng ký

Tham gia cộng đồng Xoai Aura để trải nghiệm sự cân bằng tuyệt đối.

HỌ VÀ TÊN

Nguyễn Văn A (2)

EMAIL

example@xoai-aura.com (3)

MẬT KHẨU

..... (4)

XÁC NHẬN MẬT KHẨU

..... (5)

(6) ĐĂNG KÝ

Đã có tài khoản? [Đăng nhập ngay](#) (7)

ĐIỀU KHOẢN BẢO MẬT

[Content #2]

- **Description:** This interface enables new guests to establish their secure identity profile within the resort's ecosystem. The registration payload undergoes strict data integrity validations to ensure account uniqueness and robust security standards before database persistence.
- **Mapped Use Case:** UC03 - New account registration and basic profile management.

[Content #3]

Field Name	Description
Field Group: Header & Basic Information	
(1) Header & Message	Read-only text block. Displays the "Đăng ký" (Register) title and community onboarding subtext.
(2) HỌ VÀ TÊN (Full Name)	Data type: String. Status: Required.
(3) EMAIL	Data type: String. Status: Required. Frontend must enforce standard Regex validation. SQL Constraint: The backend must handle database-level UNIQUE constraints. If a duplicate exists, the API must return a standardized 400 Bad Request error.
Field Group: Security Configuration	
(4) MẬT KHẨU (Password)	Data type: String. Input is masked by default. Incorporates an interactive eye icon to toggle visibility state (plaintext/masked).
(5) XÁC NHẬN MẬT KHẨU (Confirm Password)	Data type: String. Frontend Logic: Strict real-time validation is required. The string value must match field (4) exactly; otherwise, form submission is disabled.
Field Group: Actions & Navigation	
(6) "ĐĂNG KÝ" (Register) Button	Action: Submits the user creation payload to the backend API. Backend Logic: The server must execute one-way cryptographic hashing (e.g., BCrypt) on the plaintext password prior to executing the SQL INSERT statement.

(7) "Đăng nhập ngay" (Login Now) Link	Action: Navigates existing users back to the primary Authentication Screen (UC01).
---------------------------------------	--

3.2.3 Password Recovery Screens

[Content #1]

- Phase 1 (Request): Annotate (1) Header & Instructions, (2) Email Input, (3) "Gửi yêu cầu" Button, (4) "Quay lại đăng nhập" Link.
- Phase 2 (Reset): Annotate (5) Header & Instructions, (6) New Password Input, (7) Confirm Password Input, (8) "Cập nhật mật khẩu" Button, (9) "Quay lại đăng nhập" Link.
- The UI maintains visual consistency with the Authentication module, utilizing centered cards, elegant serif typography, and minimalist input fields.

Khôi phục mật khẩu ⁽¹⁾

Nhập email của bạn để nhận liên kết thiết lập lại mật khẩu.

Email

name@example.com ⁽²⁾

Gửi yêu cầu ⁽³⁾

⁽⁴⁾
[← Quay lại đăng nhập](#)

Thiết lập mật khẩu mới (5)

Vui lòng nhập mật khẩu mới cho tài khoản của bạn.

Mật khẩu mới

..... (6) 

Xác nhận mật khẩu mới

..... (7) 

(8) Cập nhật mật khẩu

(9) ← Quay lại đăng nhập

[Content #2:]

- Description: This feature provides a secure, two-step workflow for account recovery. The system never exposes original credentials; instead, it dispatches a secure, single-use token via email (Screen 1), authorizing the user to define a new password (Screen 2).
- Mapped Use Case: UC04 - Forgot password, reset password, and security configuration.

[Content #3]

Field Name	Description
Field Group: Phase 1 - Request Reset Link	

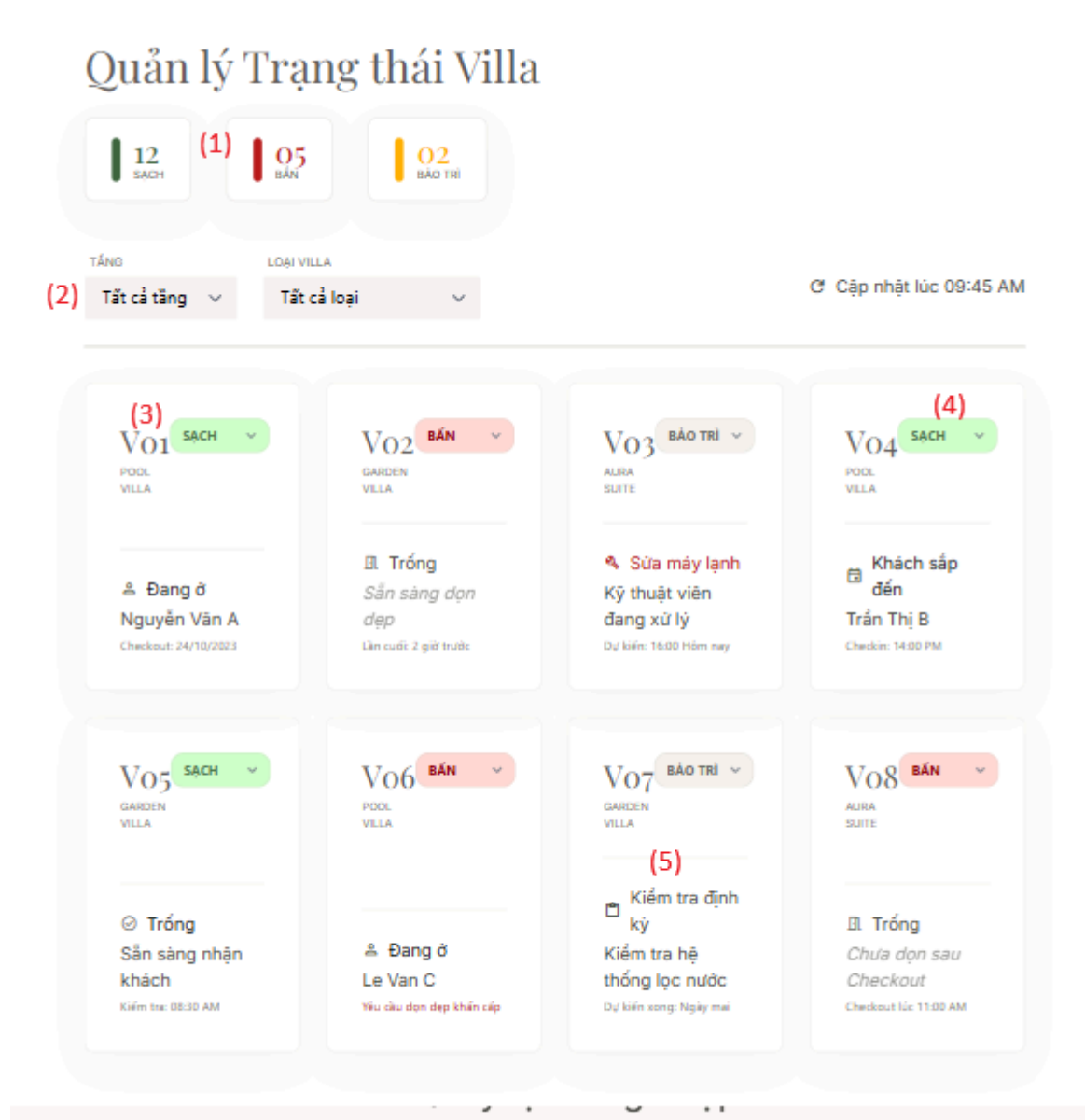
(1) Header & Instructions	Read-only text. Displays "Khôi phục mật khẩu" (Password Recovery) and instructions.
(2) Email Input	Data type: String. Status: Required field.
(3) "Gửi yêu cầu" (Send Request) Button	Action: Submits the recovery request to the API. Core Backend Logic: The system verifies the email existence. If valid, it generates a cryptographically secure Reset-Token (with a strict expiration, e.g., 15 minutes) and triggers an asynchronous email service containing the tokenized URI.
(4) "Quay lại..." (Back to Login) Link	Action: Navigates back to the Login Screen (UC01).
Field Group: Phase 2 - Set New Password	
(5) Header & Instructions	Read-only text. Displays "Thiết lập mật khẩu mới" (Set New Password). This interface is exclusively accessed via the emailed URI.
(6) Mật khẩu mới (New Password)	Data type: String. Masked input with a toggle visibility icon. Enforces strict Password Policy configurations.
(7) Xác nhận mật khẩu (Confirm Password)	Data type: String. Masked input. Frontend Logic: Must exactly match field (6) before form submission is enabled.
(8) "Cập nhật..." (Update Password) Button	Action: Submits the new password payload alongside the URL-extracted Token. Database Security: The backend validates the Token's integrity. Upon success, it executes a secure hash of the new password, updates the record, and immediately revokes the Token to prevent Replay Attacks.
(9) "Quay lại..." (Back to Login) Link	Action: Navigates back to the Login Screen (UC01).

3.3 Master Data

3.3.1 Villa Status Management Screen

[Content #1]

- The interface employs a modern Dashboard layout utilizing interactive Cards to represent individual Villas.
- The top section highlights aggregate KPI counters for room states (Clean, Dirty, Maintenance) utilizing strict traffic-light color coding. The underlying grid renders detailed Villa cards that seamlessly merge interactive state mutation capabilities (dropdowns) with cross-departmental occupancy data (Guest details, timelines).



[Content #2]

System Administration

- **Description:** This operational hub synchronizes workflows between the Front Desk and Housekeeping departments. By providing real-time physical status monitoring, the system acts as a strict guardrail, completely preventing operational anomalies such as allocating arriving guests to unserviced or out-of-order accommodations.
- **Mapped Use Case:** UC09 - Manage and update the physical status of Villas (Vacant, Occupied, Dirty/Needs Cleaning, Maintenance).

[Content #3]

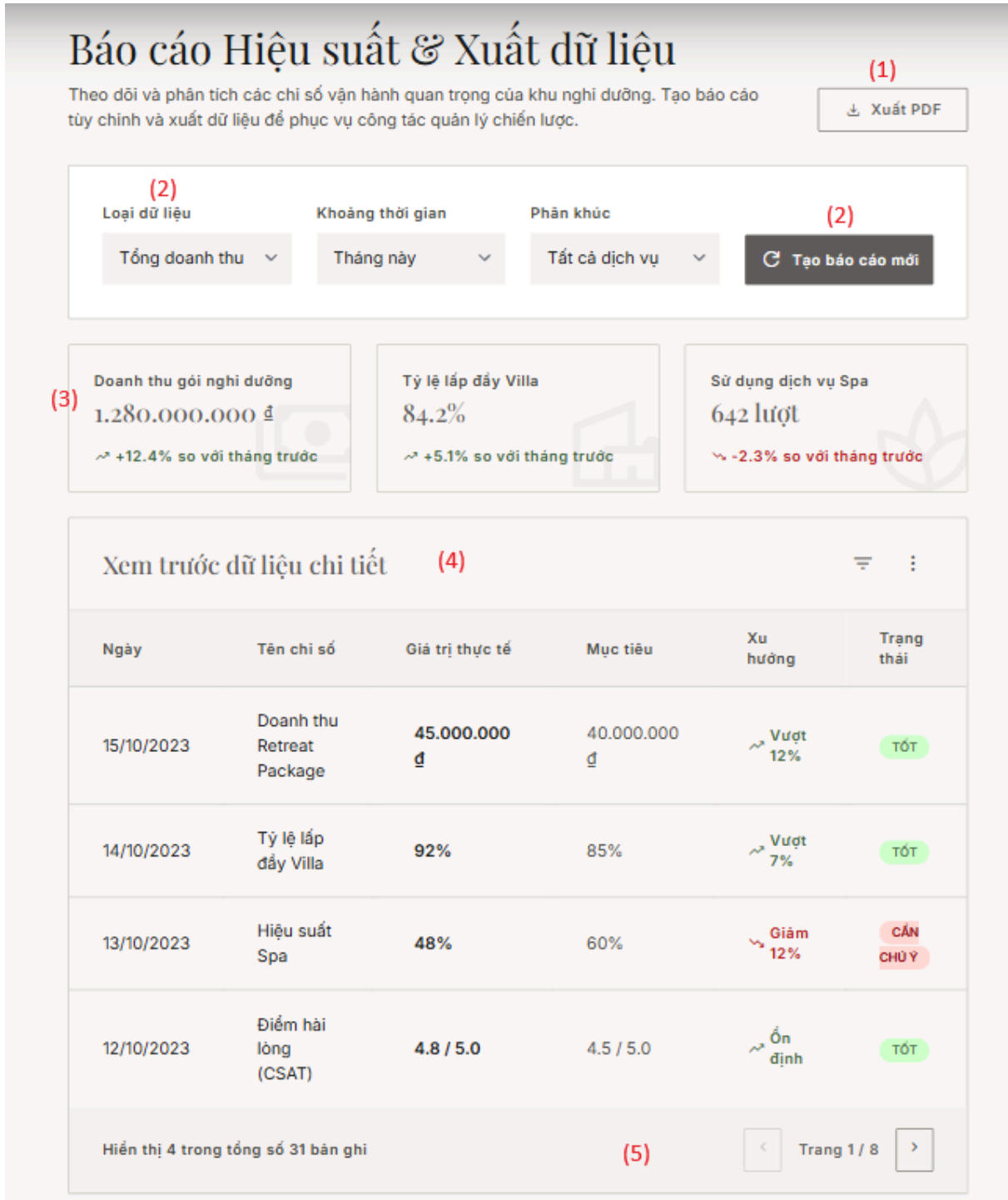
Field Description

Field Name	Description
Field Group: Dashboard Overview & Filters	
(1) KPI Status Summary	IRead-only dynamic metrics. The backend executes COUNT queries to dynamically aggregate the volume of Villas in SẠCH (Clean), BẨN (Dirty), and BẢO TRÌ (Maintenance) states.
(2) Filters	Data type: Dropdowns. Enables list segmentation by Floor (TẦNG) or Villa Category (LOẠI VILLA). Mutating these parameters instantly triggers a backend API request to re-fetch the grid payload.
Field Group: Villa Card Details	
(3) Villa Identification	Read-only text. Displays the primary key identifier (e.g., V01) and associated Room Category.
(4) Housekeeping Status Dropdown	Data type: Enum. Logic: Empowers staff to mutate the physical state directly on the card. Transitioning a state from BẨN (Dirty) to SẠCH (Clean) commits the update to the DB and concurrently dispatches an event to unlock the room's availability for Front Desk Check-in operations.
(5) Occupancy Info & Alerts	Dynamic data block requiring cross-module relational queries (e.g., JOIN with Room_Booking). Business Rules: - Active Booking: Renders "Đang ở" (Occupied), Guest Name, and Checkout timeline. - Impending Arrival: Renders "Khách sắp đến" (Arrival Expected). - No active/pending linkage: Renders "Trống" (Vacant). - Maintenance state active: Prominently renders technical fault notes (e.g., "Sửa máy lạnh" / AC Repair) in red.

3.3.2 Performance Reports & Export Screen

[Content #1]

- The interface utilizes a highly professional Admin Dashboard layout, leveraging adequate white space for optimal data readability.
- The screen is vertically divided into three functional zones: Report parameter configuration, high-level KPI summaries featuring Month-over-Month (MoM) growth indicators, and a granular, paginated data grid highlighting status alerts.



[Content #2]

- Description: This functionality provides Management with a robust analytics engine. The system automatically aggregates transactional data across all operational modules (Front Desk, Spa, F&B) to calculate room occupancy, total revenue, and service utilization rates.

System Administration

Users can preview these metrics dynamically or export them into document formats (PDF/Excel) for strategic planning.

- Mapped Use Case: UC25 - As a Manager, I want to export monthly Room Occupancy & Therapist Utilization reports to a file.

[Content #3]

Field Description

Field Name	Description
Field Group: Report Configuration & Export	
(1) "Xuất PDF" (Export PDF) Button	Action: Initiates the document generation pipeline. Core Backend Logic: The server utilizes the currently filtered dataset and applies a document rendering library (e.g., iTextPDF in Java) to construct a branded table format. It returns a Byte Stream prompting an automatic .pdf download in the browser.
(2) "Tạo báo cáo mới" (Generate Report) Button	Action: Dispatches a GET request with the active parameters to the backend, triggering a recalculation and re-rendering of both the KPI Cards and the Data Grid below.
(2) Data Filters	Data type: Dropdown Lists. - Loại dữ liệu (Data Type): e.g., Total Revenue, Spa Efficiency, Occupancy Rate. - Khoảng thời gian (Timeframe): This Month, This Quarter, This Year. - Phân khúc (Segment): All Services, Retreat Packages, A-la-carte. Logic: These selections formulate the exact parameter payload sent to the backend SQL queries.
Field Group: Performance Dashboard	
(3) KPI Summary Cards	Read-only dynamic widgets. Displays core metrics: Revenue, Occupancy Rate (%), and Spa Utilization Count. Calculation Logic: Incorporates a Trend Analysis indicator comparing current data against the previous month (MoM). Green denotes positive growth; Red denotes decline.
Field Group: Detailed Data Grid	
(4) Data Preview Grid	Read-only Data Grid rendering granular record details.

	Columns include: Date, Metric Name, Actual Value, Target, Trend, and Status. The "Trạng thái" (Status) column employs color-coded badging (e.g., TỐT/Good in Green, CẦN CHÚ Ý/Needs Attention in Red) for rapid cognitive processing.
(5) Pagination Controls	Navigation component. Logic: Implements Server-side pagination (utilizing SQL Limit/Offset) to ensure memory optimization and performance when handling large datasets. Displays total record counts and current page location (e.g., "Page 1 / 8").

4. Non-Functional Requirements - Đặc

4.1 External Interfaces

[This section provides information to ensure that the system will communicate properly with users and with external hardware or software/system elements.]

ID	Interface	Requirements
EI-01	User Interface	The system shall provide responsive web-based interfaces for desktop and tablet devices.
EI-02	Navigation	

		Screen flow shall follow Left → Right and Top → Bottom principles.
EI-03	Consent Interface	Sensitive health-related screens shall display consent notices before data collection.
EI-04	Dashboard Interface	Dashboard shall support filtering, sorting, searching, and exporting.
EI-05	Validation Interface	Input validation messages shall appear immediately.
EI-06	System Messages	System shall follow predefined messages MSG-01 → MSG-15.
EI-07	Payment Gateway	Support deposit payment and final payment integration.
EI-08	Notification Service	Support email verification and appointment reminders.
EI-09	Reporting Interface	Support exporting reports and invoice generation.

4.2 Quality Attributes

[List all the required system characteristics (quality attributes) specification. Some of the possible attributes are provided with the guide/descriptions are mentioned here]

4.2.1 Usability

[This section includes all those requirements that affect usability. For example, specify the required training time for a normal user and a power user to become productive at particular operations specify measurable task times for typical tasks or base the new system's usability requirements on other systems that the users know and like specify requirement to conform to common usability standards, such as IBM's CUA standards Microsoft's GUI standards]

ID	Requirement	Measurement
US-01	Guest registration shall be simple and intuitive	≤ 3 minutes
US-02	Retreat package booking shall be completed efficiently	≤ 5 minutes
US-03	Receptionist shall complete check-in quickly	≤ 3 minutes
US-04	Therapist shall update treatment status efficiently	≤ 30 seconds
US-05	Chef shall locate dietary information quickly	≤ 15 seconds
US-06	Manager shall access reports efficiently	≤ 10 seconds
US-07	User interfaces shall remain visually consistent	Across all modules
US-08	Consent requirements shall be clearly displayed	Mandatory

4.2.1.1 Training Requirements

User Role	Training Time
-----------	---------------

System Administration

Guest	No training required
Receptionist	≤ 2 hours
Therapist	≤ 1 hour
Administrator	≤ 4 hours

4.2.1.2 Usability Standards

Standard Type	Requirement
Accessibility	WCAG principles
Responsive Design	Supported
Navigation Consistency	Required

4.2.2.1 Performance

[The system's performance characteristics are outlined in this section. Include specific response times.

Where applicable, reference related Use Cases by name.

Response time for a transaction (average, maximum)

Throughput, for example, transactions per second

Capacity, for example, the number of customers or transactions the system can accommodate

Resource utilization, such as memory, disk, communications, and so forth.]

ID	Requirement	Target
PF-01	Average response time	≤ 2 seconds

System Administration

PF-02	Maximum response time	≤ 5 seconds
PF-03	Booking confirmation (UC07)	≤ 10 seconds
PF-04	Final payment processing (UC22)	≤ 15 seconds
PF-05	Revenue dashboard loading (UC24)	≤ 8 seconds
PF-06	Concurrent active users	100 users
PF-07	Booking requests	30 requests/minute
PF-08	Dashboard requests	50 requests/minute
PF-09	Guest Accounts	100,000
PF-10	Bookings	50,000
PF-11	Audit Logs	1,000,000
PF-12	Payment Records	500,000
PF-13	CPU Utilization	≤ 70%
PF-14	Memory Utilization	≤ 8 GB
PF-15	Database Query	≤ 1 second
PF-16	Encryption Overhead	≤ 10%
PF-17	Monthly System Uptime	≥ 99.5%

System Administration

PF-18	Recovery Time	≤ 30 minutes
PF-19	Transaction Data Loss	Not allowed

4.2.2.2 Related Business Rules

Rule ID	Description
BR-09	Sensitive Data Encryption
BR-15	Audit Trail

5. Requirement Appendix - My

5.1 Business Rules:

ID	Business Rule Name	Detailed Description (Constraints & Logic)	Reference (UC/Source)
BR-01	Booking Confirmation and Deposit Payment	A Retreat Package booking shall only be confirmed after the system receives a successful deposit payment result from the payment gateway.	UC07
BR-02	Villa Management and Allocation	Guests may only select a Villa Type when making a booking. The system shall only confirm a booking if sufficient capacity is available for the selected Villa Type during the requested period. A specific Villa shall be assigned by the Receptionist during the Check-in process based on actual Villa availability.	UC07, UC08
BR-03	Villa Status and Allocation Constraints	A Villa shall not be assigned to more than one active booking during the same period. Villas with a status of Maintenance or Out of Service shall not be allocated to guests.	UC08, UC09

BR-04	Dual Resource Spa Scheduling	A Spa appointment shall only be valid when both an available Therapist and an available Therapy Room exist at the requested time. The system shall prevent double booking of resources.	UC12
BR-05	Spa Service Eligibility and Update Control	Guests may only book Spa services included in their purchased Retreat Package. Only the assigned Therapist may update the treatment session status. Additional Spa services outside the package may only be added by a Receptionist and must be recorded in the Guest Folio.	UC11, UC14, UC15
BR-06	Automatic F&B Menu Filtering	The system shall automatically exclude menu items that contain allergens or conflict with the guest's declared dietary preferences. Chefs and F&B Staff shall only have access to allergy and dietary information necessary for their work.	UC16, UC17, UC20
BR-07	Role-Based Access Control (RBAC) and Data Minimization	Therapists may only access health information required for treatment purposes. Chefs may only access food allergy and dietary preference information. Receptionists shall not have access to guest health records. Access control shall be enforced at the backend level.	UC03, UC13, UC17, UC20
BR-08	Consent for Sensitive Data Collection	The system shall obtain explicit consent from guests before collecting health-related or allergy information. Consent checkboxes shall be unchecked by default.	UC02
BR-09	Sensitive Data Encryption	Health information, allergies, dietary preferences, and personal identification data shall be encrypted when stored in the database.	UC02, UC08
BR-10	Right to Data Erasure	Guests may request the permanent deletion of their health and allergy information after their retreat stay has been completed.	UC05
BR-11	Guest Folio and Consolidated Billing	All Spa and F&B charges shall be recorded in the Guest Folio using the corresponding Booking_ID and included in the final bill.	UC15, UC19, UC21

System Administration

BR-12	Check-out Constraints	Guests shall not be allowed to complete the Check-out process if any Spa or F&B charges remain unpaid. Any previously paid deposit shall be deducted from the final invoice.	UC21, UC22
BR-13	Reporting and Review Logic	Revenue and occupancy reports shall only include completed transactions. Only guests who have completed their retreat stay may submit reviews and ratings. Each booking may submit only one review.	UC23, UC24, UC25
BR-14	Guest Stay Registration Information	During Check-in, the system shall collect and store guest identification information to comply with accommodation registration regulations.	UC08, Residence Law 2020
BR-15	Audit Trail Management	The system shall maintain audit logs for critical activities such as login, health data access, booking, payment, and Check-out to support monitoring and traceability.	UC07, UC11, UC21, UC22
BR-16	Meal Order Status Workflow	Meal Order status shall only progress in the following sequence: Pending → Preparing → Ready for Delivery. Status reversal shall not be permitted. Only Chefs or F&B Staff may update Meal Order status.	UC18
BR-17	Spa Appointment Notification and Synchronization	After a Spa appointment is successfully booked, the system shall send confirmation and reminder notifications to the guest. Notification failures shall not invalidate a confirmed appointment.	UC11
BR-18	Authentication and Single Sign-On (SSO)	The system shall support authentication through Google and Facebook. Accounts registered via SSO must complete email verification before being allowed to book a Retreat Package. The system shall prevent duplicate account creation.	UC01

5.2 System Messages

#	Code	Type	Context	Message
1	MSG-01	Success	Account registration successful	Registration successful. Please verify your email address.
2	MSG-02	Error	Login failed	Invalid email or password.
3	MSG-03	Success	Email verification successful	Your email has been verified. You may now log in.
4	MSG-04	Warning	Health data consent not provided	You must provide consent before submitting health information.
5	MSG-05	Success	Retreat Package booking successful	Retreat Package booking created successfully.
6	MSG-06	Error	Deposit payment failed	Deposit payment failed. Please try again.
7	MSG-07	Success	Check-in successful	Check-in completed successfully.
8	MSG-08	Error	No suitable Villa available	No available Villa could be found for allocation.
9	MSG-09	Success	Spa appointment booked successfully	Spa appointment booked successfully.
10	MSG-10	Error	Therapist or therapy room unavailable	No available Therapist or Therapy Room could be found.
11	MSG-11	Success	Meal order placed successfully	Meal order has been recorded successfully.
12	MSG-12	Warning	Meal contains allergens	Warning: This meal contains ingredients listed in the guest's allergy profile.
13	MSG-13	Warning	Outstanding charges during check-out	Please settle all outstanding charges before checking out.

System Administration

14	MSG-14	Success	Check-out completed successfully	Check-out completed successfully.
15	MSG-15	Success	Health data deleted successfully	Health data has been deleted successfully.
16	MSG-16	Success	Review submitted successfully	Thank you for submitting your review.
17	MSG-17	Success	Excel report exported successfully	Report exported successfully.
18	MSG-18	Error	Unauthorized access	You do not have permission to access this function.
19	MSG-19	Error	Unexpected system error	An unexpected system error has occurred. Please try again later.